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Center for Multiscale Modeling of Multiphase Combustion
University of Florida

PSAAP4 Semi-Annual Report

June 1, 2026

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2 ACCOMPLISHMENTS

2.1 Overall Objectives

The University of Florida (UF), in collaboration with the Georgia Institute of Technology (GT) and Purdue University (PU), has established a Center for Multiscale Modeling of Multiphase Combustion (CM³C) as part of the NNSA’s Predictive Science Academic Alliance Program-IV’s broad single science/engineering-focused Predictive Simulation Center (PSC). We plan to solve the problem of phase change and group combustion of a dense cloud of particles under high-speed and detonation conditions at an unprecedented level of physical detail with a state-of-the-art multiscale modeling framework that combines the powers of scientific computing (SC) and machine learning (ML). The key technical objective of the center is to predict the spatio-temporal higher-order statistics of phase change and combustion in polydisperse multiphase systems in high-speed environments through exascale computation. The intellectual objective of the work is to understand group combustion via a multiscale approach, where the micro-meso-macroscales are linked by hierarchical physics-informed ML models that will enable truly predictive large-scale simulations.

The group combustion of a cloud of particles in a high-speed environment is a formidable problem, as it combines turbulence, multiphase flow, shock waves, and reactive flow physics. The wide range of scales, phase changes, and stiff chemical reactions make direct numerical simulations (DNS) of combustion of even an isolated particle in a high-speed environment a problem worthy of exascale computation. The collective burning of a cloud of particles cannot be approached as a superposition of individual burning processes. Particle-gas interactions at the mesoscale through preferential fingering, particle-gas-particle interactions at the microscale in the form of localized oxidizer depletion in reacting wakes, and close-range particle-particle interactions through collisions render the problem far more complex, and multiscale methods are required for efficient use of exascale facilities.

We will develop a new hierarchical scale-bridging approach for high-fidelity closure models by providing proper context to the bottom-up approach currently employed in complex multiscale multiphase problems. Instead of building closure models based only on the canonical configuration of an isolated burning particle, we inform the coarse-grained models with a complete statistical description of all possible group combustion configurations. To achieve this goal, we leverage advancements in AI, melding SC with ML at multiple scales. We advance a co-design strategy that combines particle-resolved (PR), Euler-Lagrange (EL), and Euler-Euler (EE) simulations of high-speed combustion of dense particle clouds with physics-informed generative and operator-learning models.

2.1.1 Center Roadmap

The roadmap of the center is captured in fig. 1. The figure shows how the major center activities are integrated into achieving the overarching goals of the center. The key activities of the center are

- Box 1B: High-fidelity particle-resolved (HF-PR) direct numerical simulations of $\mathcal{O}(1000)$ burning carbon particles in high-speed cross-flow with detailed chemistry and compressible flow features.
- Boxes 1C and 1D: Generative machine learning (GML) model that learns from the HF-PR simulations. Once trained the GML can serve as lower-fidelity surrogate and produce particle-

resolved information of burning carbon particles in high-speed cross-flow at order of magnitude lower computational cost, which preserving adequate accuracy.

- Box 1E: Operator learning machine learning (OLML) uses the HF-PR and the lower-fidelity surrogate data in a multi-fidelity framework to develop best-possible models of all the closure terms needed in the coarse-grained Euler-Lagrange (EL) large eddy simulations (LES).
- Box 2B: Perform mesoscale EL-LES simulations of a curtain of reactive particles being subjected to curved shock. Validate the results against mesoscale experiments being planned at the Sandia multiphase shock-tube facility.
- Box 3A: Perform macroscale EL-LES simulations of explosive dispersal of reactive particles. Validate the results against macroscale experiments being performed at Purdue.
- Boxes 3B and 3C: Use the meso and macroscale EL-LES simulation results to develop additional closure models needed for Euler-Euler (EE) simulations and perform EE-LES macroscale simulations of even larger systems containing trillions of reactive particles.
- Feedback loop: Rigorous validations will be performed at all three scales. All contributing uncertainties will be propagated and quantified. A detailed budget of the contributing uncertainties will identify the best candidates for uncertainty reduction. Accordingly, the next cycle of HF-PR microscale simulations and meso/macroscale simulations will be performed. Through active learning the machine learning models will also be improved in each cycle of iteration.

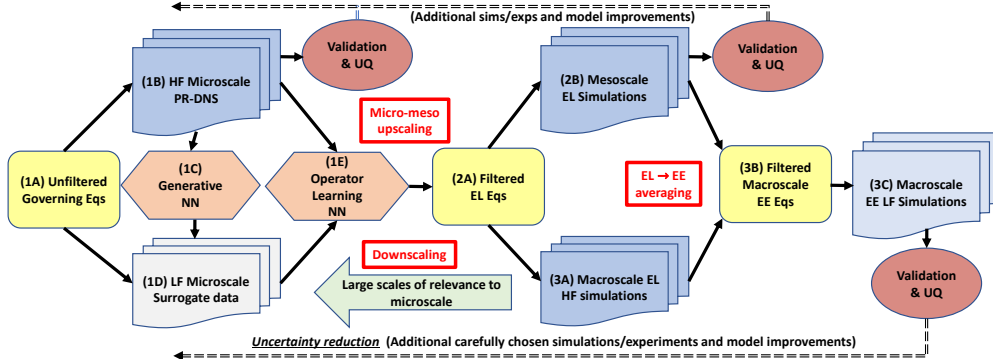


Figure 1: Integration of operator-learning and generative neural networks with scale-separated exascale simulations. HF/LF: high/low-fidelity, UQ: uncertainty quantification, EL: Euler-Lagrange, EE: Euler-Euler, NN: neural network.

Exascale Simulations: Exascale simulations will be performed at all three scales using the 2025 Gordon-Bell Finalist code, Multicomponent Flow Code (MFC). High Fidelity Particle-resolved (HF-PR) simulations will be performed at the microscale by solving the complete gas and particulate phase governing equations (block 1A of fig. 1). The heterogeneous surface and gas-phase reactions will be considered with the optimal reduced-order reactions. A set of simulations will consider stationary carbon rods. With the uncertainties of reduced-order chemistry established, microscale simulations will be performed for the reactive C and Mg particles (block 1B of fig. 1). Mesoscale EL simulations of the two configurations presented in fig. 2(b,c) will solve the filtered governing equations (block 2A). The subgrid closure models will initially be the best currently available. From year 3 onward, the EL simulations will employ the new ML-based closure models (block 1E). We will perform mesoscale simulations (block 2B) and validate against experimental results.

We will perform two types of macroscale simulations. The largest possible hero simulations of the demonstration problem will be performed yearly with increased computational resources and improved modeling fidelity. These EL simulations (block 3A) will employ the ML-based best-possible closure models *relayed* from the mesoscale. We will also conduct many low-fidelity EE simulations (block 3C) by solving the filtered gas and particulate phase governing equations (block 3B) [Balachandar, 2024a]. Ensemble averaging the ML-based EL closure models (block 1E) over appropriate particle distributions will obtain the best possible EE closure models.

In addition to simulating the demonstration problem, we will perform increasingly complex simulations at the micro and mesoscales. The ML models developed from these micro/mesoscale simulations will be used in Y3–Y5 macroscale simulations. The proposed validation experiments at the micro and mesoscale are shown in fig. 2. To take advantage of validation experiments, many low-fidelity Euler-Euler simulations will be used for calibration.

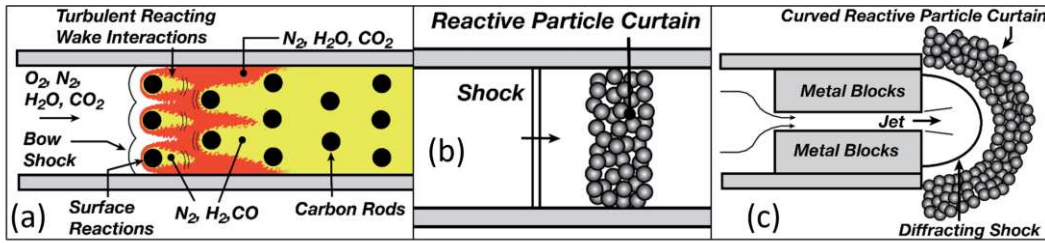


Figure 2: Schematic of the (a) microscale, (b) planar mesoscale, and (c) cylindrical mesoscale configurations. Simulations will employ particle-resolved (PR) methodology at the microscale and Euler–Lagrange (EL) methodology at the mesoscale.

2.2 What was accomplished under these goals?

	Y1	Y2	Y3	Y4	Y5		Y1	Y2	Y3	Y4	Y5
Simulations						Experiments					
HF-PR Micro (C rods)						Micro C rods					
HF-PR Micro (C & Mg)						Meso Campaigns					
EL Mesoscale						Macro (PU)					
EL Macroscale						Macro (other)					
ML development						CS/Exascale					
Generative ML						APU (El Capitan)					
Surrogate data						Dyn. Load Bal.					
OMLM deterministic						Integr. ML-closure					
Mori Zwanzig						Trace-based					
VVUQ						Software					
Relay validation						Software upgrades					
UQ micro						Regression testing					
UP across scales						Release - Minor					
UB and UR						Release - Major					

Table 1: Gantt chart for CM³C that shows the timeline of all the important activities of the center.

The expected 5-year timeline of the different activities of the center are shown in Table 1. During the past six months the center has progressed in accordance with the plan outlined in the Table. The Key accomplishments of the center since its establishment are as follows. They are directed by the center roadmap with the focus towards (i) building the modeling and exascale simulation

capabilities needed, (ii) periodically performing the planned micro/meso/macro simulations, and (iii) rigorous validation against companion micro/meso/macro experiments, followed by uncertainty quantification. Each of the accomplishment towards these goals will be elaborated in greater detail in the following sections.

- Recruitment of top notch talent and engaging them in interacting teams.
- Getting the exascale simulation framework, Multicomponent Flow Solver (MFC), ready to perform the proposed HF-PR microscale simulations.
- Getting the exascale simulation framework, Multicomponent Flow Solver (MFC), ready to perform the proposed EL LES simulations of explosive dispersal of reactive particles.
- Establishing the framework and workflow pipeline for generative ML modeling of reactive particle-laden flows.
- Identifying all the closure models that need to be considered in the operator learning framework for compressible reactive multiphase flow and develop the graph neural network pipeline.
- Developing the group combustion modeling theory upon which the “best-possible” machine learning closure models will be constructed. Such model aims to deterministically account for neighbor-informed burning of a swarm of particles/droplets.
- Identifying all the sources of uncertainty in HF-PR simulations, generative and operator-learning models, and EL LS simulations. Establishing the experimental and measurement processing uncertainties for the carbon rod experiments.

2.2.1 Recruitment

The Center for Multiscale Modeling of Multiphase Combustion has successfully recruited truly outstanding PhD students and postdocs in all areas of the proposed work. These include students in fluid mechanics, combustion, machine learning, and computational science. See Section 7 for a complete list of students and postdocs funded by the Center.

In addition, we have also incorporated outstanding leveraged PhD students and postdocs. Leveraged students are those individuals who are not directly funded by the Center resources, but are highly interested in the scientific questions and models that are being addressed and developed by the Center. Their participation into the daily activities of the Center have already made a major impact on the successes so far.

These students are grouped into the following teams:

1. Microscale Team - responsible for carrying out particle-resolved simulations of thousands of reacting particles for ML training and ML model development. Sec. 2.2.2 will discuss their progress.
2. Meso/Macroscale Team - responsible for carrying out Euler-Lagrange EL simulations at the meso and macroscales, including the demonstration problem. Sec. 2.2.3 will discuss their progress.
3. MFC code development team - responsible for integrating and updating MFC according to the required physics needed at all three scales (micro, meso, macro). Sec. 2.2.4 will discuss their progress.

4. Experimental Team - responsible for carrying out validation-quality experiments at the microscale (carbon rod experiments) and the mesoscale (hemispherical explosive surrounded by a bed of particles). Sec. 2.2.5 will discuss their progress.
5. Machine Learning Team - responsible for developing next-generation models (both generative NN and operator learning NN) for group combustion. Sec. 2.2.6 will discuss their progress.
6. Group Combustion Team - responsible for developing deeper understanding of the interaction of reacting droplets or particles, assisting the Microscale Team in physics for PRS, the Macroscale Team in developing appropriate EL models for reacting particles, and assisting the ML for model development. Sec. 2.2.7 discusses their progress.
7. UQ/Workflow Team - responsible for evaluating the uncertainty of each of the modules being integrated and propagating the uncertainty to predict the overall uncertainty. This team will also perform uncertainty budget to identify the major contributors to uncertainty which will drive the subsequent uncertainty reduction steps. Integrating the various efforts of the center and ensuring proper work/data flow is also this team's responsibility.

The postdocs oversee and participate with one or more teams, and are responsible for leading the weekly team meetings. The weekly team meetings are cross-populated with appropriate members from the other teams and faculty.

2.2.2 HF-PR Microscale Simulations with MFC

The microscale team has made an astonishing amount of progress in the first six months of the project on getting MFC ready for particle-resolved simulations (PRS) for group combustion; see table 2. We recall that the purpose of the particle-resolved microscale simulations is to use the simulation results in the development of generative and operator-learning ML models. To train these models we must first identify all the quantities that must be calculated in the HF-PR simulations. In this section we highlight (i) the closure terms that the ML team will model using the data obtained from the PR simulations; (ii) the spatial filtering operation that is currently being implemented into MFC; and (iii) selected cases produced by the microscale team.

Requirement	Previous Capability	Current Capability
Solid-to-Fluid Force Spreading: MFC must be able to handle force spreading onto the fluid at the IB interface point	Already Supported in MFC	Already Supported in MFC
Fluid-to-Solid Force Spreading: MFC must be able to compute and apply forces on immersed boundaries from the fluid	Not Yet Implemented	Fully Implemented
Immersed Boundary Motion: IBs need to be able to move through the fluid	Not Yet Implemented	Fully Implemented
IB-IB Collision Forces: Resolved particles need to be able to collide with each other and the walls of the simulation	Not Yet Implemented	Fully Implemented
IB-EL Collision Forces: Resolved particles must be able to collide with sub-grid particles	Not Yet Implemented	Will collaborate with EL implementation
Scaling: MFC must be able to run up to 1000 moving resolved particles at once	MFC Could Run Up to 10 Particles	Fully Implemented
Reacting Boundaries: MFC needs to support several boundary conditions on IB surfaces for chemistry and temperature changes	Not Yet Implemented	Not Yet Implemented

Table 2: List of tasks accomplished by the microscale team.

(A) EL formulation and Closure Terms: Our goal is to understand and develop models for all the closure terms that can be used at the meso and macroscale simulations. This information will also be used by the ML Team to develop comprehensive operators for the closure terms. Below is an outline of the equations and their definitions. The viscous, compressible equations that are solved at the meso and macroscales are given by

$$\frac{\partial}{\partial t} (\phi_g \bar{\rho}_g) + \nabla \cdot (\phi_g \bar{\rho}_g \tilde{\mathbf{u}}_g) = 0 \quad (2.1)$$

$$\frac{\partial}{\partial t} (\phi_g \bar{\rho}_g \tilde{\mathbf{u}}_g) + \nabla \cdot (\phi_g \bar{\rho}_g \tilde{\mathbf{u}}_g \otimes \tilde{\mathbf{u}}_g) = -\phi_g \nabla \bar{p}_g + \boxed{\phi_g \nabla \cdot \bar{\boldsymbol{\tau}}_g} - \nabla \cdot (\phi_g \mathbf{R}_{sg}) - \mathbf{F}'_g \quad (2.2)$$

$$\begin{aligned} \frac{\partial}{\partial t} (\phi_g \bar{\rho}_g \tilde{E}_g) + \nabla \cdot (\phi_g \bar{\rho}_g \tilde{\mathbf{u}}_g \tilde{E}_g) = & -\nabla \cdot (\phi_g \bar{p}_g \tilde{\mathbf{u}}_g) + \boxed{\nabla \cdot (\phi_g \bar{\boldsymbol{\tau}}_g \cdot \tilde{\mathbf{u}}_g)} \\ & - \bar{p}_g \nabla \cdot (\phi_p \mathbf{u}_p) + \boxed{\bar{\boldsymbol{\tau}}_g : \nabla (\phi_p \mathbf{u}_p) - \phi_g \nabla \cdot \bar{\mathbf{q}}_g} - (W'_g + Q'_g) - \nabla \cdot (\phi_g \mathbf{Q}_{sg}). \end{aligned} \quad (2.3)$$

Here, overbar indicates phase average, and therefore $\bar{\rho}_g$, $\bar{p}_g$, $\bar{\boldsymbol{\tau}}_g$, and $\bar{\mathbf{q}}_g$ are gas density, pressure, viscous stress, and heat flux averaged over the gas phase. On the other hand, $\bar{u}_p$ is particle velocity averaged over the particulate phase. In essence, overbar quantities with subscript g are averaged over the gas phase, while overbar quantities with subscript p are averaged over the particle phase. Tilde denotes FabRe-averaged quantities and these are used only in the gas phase. Thus, $\tilde{\mathbf{u}}_g$ and $\tilde{E}_g$ denote FabRe-averaged gas phase velocity and total energy. The blue highlighted terms are viscous/diffusive in origin and they are neglected if the gas phase is treated inviscid.

The terms $\mathbf{F}'_g$, W'_g , and Q'_g are the feedback terms due to the presence of the particles and they do not include the undisturbed flow components. They are treated as sum over all the N_p particles. For the force we have

$$\mathbf{F}'_g(\mathbf{x}, t) = \sum_{i=1}^{N_p} \int_{S_i} G(\mathbf{x} - \mathbf{x}') \mathbf{F}'_{gp,i}(\mathbf{x}', t) dS \approx \sum_{i=1}^{N_p} G(\mathbf{x} - \mathbf{x}_i) \mathbf{F}'_{gp,i} \quad (2.4)$$

where the integral is over the surface of the i^{th} particle and $\mathbf{F}'_{gp,i}$ is the perturbation fluid dynamic tractional force distribution around the surface of the particle. Here G is the filter function usually taken to be a Gaussian. If the length scale of the filter function is much larger than the size of the particle, G does not vary substantially over the surface of the particle, and therefore can be taken out of the integral. This leads to the approximation where $\mathbf{F}'_{gp,i} = \int_{S_i} \mathbf{F}'_{gp,i} dS$. Similar expressions and approximations apply for W'_g and Q'_g . Note that these terms are defined positive for the particles, and therefore they appear with the negative sign in the gas phase equations.

The perturbation force, work, and heat transfer terms of the individual particles, i.e., $\mathbf{F}'_{gp,i}$, $W'_{gp,i}$, and $Q'_{gp,i}$ for the *interphase closure* terms and they must be modeled. There have been substantial effort directed towards their modeling. For detail see Daoud *et al.* (2026). In this document we will focus more on modeling of pseudo turbulence, since far less attention has been paid to it in the past. But the modeling strategy to be elaborated below for pseudo turbulence applies for modeling of interphase closure as well.

(B) Spatial Filter Operation in MFC: We must build a three-dimensional filter operation so that from any PRDNS field $\psi(\mathbf{x})$ we can calculate the filtered field $\bar{\psi}(\mathbf{x}')$ using the convolution operation

$$\bar{\psi}(\mathbf{x}') = \iiint \psi(\mathbf{x}) G(\mathbf{x}, \mathbf{x}'; \mathcal{L}_f) d\mathbf{x}', \quad (2.5)$$

where $G = G_x G_y G_z$ is the 3D filter function. It can be written as a product of three 1D filter functions, each of which is characterized by the length scale $\mathcal{L}_f$. While the PRDNS grid is typically much finer than a particle, the filter width $\mathcal{L}_f$ will be chosen to be of the order of the particle size. Thus, the filtered quantity $\bar{\psi}(\mathbf{x}')$ can be output on a much coarser grid. The filter operation, however, requires long-distance communication. I.e., $\bar{\psi}$ at a point $\mathbf{x}'$ requires weighted averaging of ψ at grid points that are $O(\mathcal{L}_f)$ away. There are clever ways to implement this, which will be discussed next.

The convolution integrals given in (2.5) can be discretized and the discrete filter operation can be expressed as

$$\bar{\psi}(x_I, y_J, z_K) = \sum_{k=-L_z}^{L_z} W_k \left(\sum_{j=-L_y}^{L_y} W_j \left(\sum_{i=-L_x}^{L_x} W_i \psi(x_{I+i}, y_{J+j}, z_{K+k}) \right) \right), \quad (2.6)$$

where W_i , W_j , and W_k are weights that implement the filter integrals. The nesting shown above is to stress the point that directional splitting applies and as a result the sum along the three coordinates can be done one after the other. Here L_x , L_y , and L_z are the **half-stencil width** along the three directions beyond which the filter function can be considered to have sufficiently decayed. The half-stencil widths depend on filter width $\mathcal{L}$ and the grid size along the three directions.

Let us now consider the filter operation only along the x -direction for all the grid points that are along that direction. The filter operation can then be expressed as

$$\begin{bmatrix} \bar{\psi}(x_1) \\ \bar{\psi}(x_2) \\ \vdots \\ \bar{\psi}(x_N) \end{bmatrix} = \mathbf{A}_x \begin{bmatrix} \psi(x_1) \\ \psi(x_2) \\ \vdots \\ \psi(x_N) \end{bmatrix}, \quad (2.7)$$

where $\mathbf{A}_x$ is the discrete filter matrix that is as large as the number of grid points along the x -direction. It is however banded with a **half width** of L_x . The impossibility of a naive implementation of the filter operation in MFC becomes clear. We can expect L_x to be $O(10)$, since it is of the same order as the number of grid points used to discretize one particle. Whereas, MFC typically uses only one or two ghost cells. Thus naive filter operation is not possible at rank boundaries.

A simple solution can be sought with the following decomposition:

$$\mathbf{A}_x = \mathbf{T}_{L_x, x} \cdots \mathbf{T}_{2, x} \mathbf{T}_{1, x} \quad (2.8)$$

where $\mathbf{T}_{1, x}$, $\mathbf{T}_{2, x}$, etc., are tri-diagonal matrices. It can be shown that the banded matrix $\mathbf{A}_x$ can be expressed as a product of L_x tri-diagonal matrices. What this means is with only one ghost fluid cell only a tri-diagonal filter operation is possible. We first do the filter operation using the tri-diagonal matrix $\mathbf{T}_{1, x}$. The output will be filtered again using the tri-diagonal matrix $\mathbf{T}_{2, x}$. By repeating this L_x times we will essentially achieve the effect of filtering with the matrix $\mathbf{A}_x$. If we employ two layers of ghost cells in MFC then the filter operation can be done with multiplication of penta-diagonal matrices as given below

$$\mathbf{A}_x = \mathbf{P}_{L_x/2, x} \cdots \mathbf{P}_{2, x} \mathbf{P}_{1, x} \quad (2.9)$$

and the number of repeated application can be reduced to $L_x/2$.

Note that none of the matrices will be really constructed. Only their action on the data will be calculated. In essence, by applying a narrow filter operation several times we can achieve the effect

of a much broader filter. This operation can be computationally demanding, but we need to perform this only infrequently whenever filtered data needs to be output.

Question 1: Given filter function (e.g., Gaussian), filter length scale (e.g., $\mathcal{L}_f = d_p/2$), grid details (e.g., Δx , Δy , and Δz), and number of ghost layers, how can the filter operation be automated?

Question 2: The filtered quantities are much smoother than the original PRDNS data. So when we write it out, we do not need to use the original MFC grid. The filtered data may require only two orders magnitude fewer grid points. A coarser grid can be used. How do we decide the coarse resolution on which to write the filtered quantities? **Answer 2:** We suggest that we choose Gaussian Filter, with filter size $\mathcal{L}_f = d_p/2$. We can then output the data at a coarser resolution of $\Delta x/d_p = 1/4$ (i.e., 4 grid points per sphere diameter).

(C) Filtered Quantities to be Computed: The following list of closure quantities are needed to develop ML models.

- Subgrid stress $\mathbf{R}_{sg} = \overline{\rho \mathbf{u} \otimes \mathbf{u}} - \frac{1}{\rho} \overline{\rho \mathbf{u}} \otimes \overline{\rho \mathbf{u}}$
- $\mathbf{Q}_T = \gamma C_v \left(\overline{\rho \mathbf{u} T} - \tilde{T} \overline{\rho \mathbf{u}} \right)$,
- $\mathbf{E}_{ku} = \frac{1}{2} \left[\overline{\rho \mathbf{u} (\mathbf{u} \cdot \mathbf{u})} - \frac{\overline{\rho \mathbf{u}}}{\rho} \overline{\rho (\mathbf{u} \cdot \mathbf{u})} \right]$
- $\mathbf{W}_{\tau u} = \frac{1}{\rho} \left(\overline{\rho \boldsymbol{\tau} \cdot \mathbf{u}} - \boldsymbol{\tau} \cdot \overline{\rho \mathbf{u}} \right)$
- $\mathbf{R}_{\mu,sg}^g = \boldsymbol{\tau} - \frac{2}{3} \mu_g (\nabla \cdot \overline{\mathbf{u}}_m) \mathbf{I} + \mu_g \left[(\nabla \overline{\mathbf{u}}_m) + (\nabla \overline{\mathbf{u}}_m)^\top \right]$, where μ_g is a function of FabRe-averaged temperature $\tilde{T} = (\overline{\rho T} / \overline{\rho})$ and the mixture velocity is defined as $\overline{\mathbf{u}}_m = \overline{\phi_g} \overline{\mathbf{u}} + \overline{\phi_p} \overline{\mathbf{u}}_p$.
- $\mathbf{R}_{\lambda,sg}^g = \overline{\mathbf{q}} - \lambda_g \nabla (\tilde{T})$, where λ_g is a function of the FabRe-averaged temperature.

By investigating these closure quantities closely, we can identify which filtered quantities are needed from the PRDNS simulations. They can be sorted into scalar vector, and second-rank tensor quantities. Thus the filtered quantities to be computed in the PRDNS are:

- **Scalar:** $\overline{\phi_p}$, $\overline{\rho}$, $\overline{\rho (\mathbf{u} \cdot \mathbf{u})}$
- **Vector:** $\overline{\mathbf{u}}_p$, $\overline{\rho \mathbf{u}}$, $\overline{\rho \mathbf{u} T}$, $\overline{\rho \mathbf{u} (\mathbf{u} \cdot \mathbf{u})}$, $\overline{\rho \boldsymbol{\tau} \cdot \mathbf{u}}$, $\overline{\mathbf{q}}$
- **Tensor:** $\overline{\rho \mathbf{u} \otimes \mathbf{u}}$, $\boldsymbol{\tau}$

Note that quantities such as $\overline{\phi_p}$ and $\overline{\mathbf{u}}_p$ must be carefully calculated using indicator function using the same filter operation defined in the previous subsection. In essence, our initial effort will aim to calculate the above 11 subgrid closure quantities rigorously from the PRDNS with the smallest filter to be considered in an EL simulation. These 11 quantities will be used to calculate $\mathbf{R}_{sg}$, $\mathbf{Q}_T$, $\mathbf{E}_{ku}$, $\mathbf{W}_{\tau u}$, $\mathbf{R}_{\mu,sg}^g$, and $\mathbf{R}_{\lambda,sg}^g$. These quantities can be further filtered with larger filters and the closure quantities evaluated at the larger filters. The closure quantities evaluated at multiple filter levels will then be the input for ML modeling.

Note: The preset approach is limited to non-reactive flows. But the framework can be easily extended to consider multiphase reactive flows, and the microscale team will do this in the coming months.

(D) PRDNS Results: Selected results highlighting the rapid progress of the Center of particle-resolved simulations using the immersed boundary method are shown in fig. 3. These results include verification studies of a single particle (top row, first two figures), capability runs of 1000 particles (top row, right), random pack of 100 spheres showing drag and lift coefficients (bottom row).

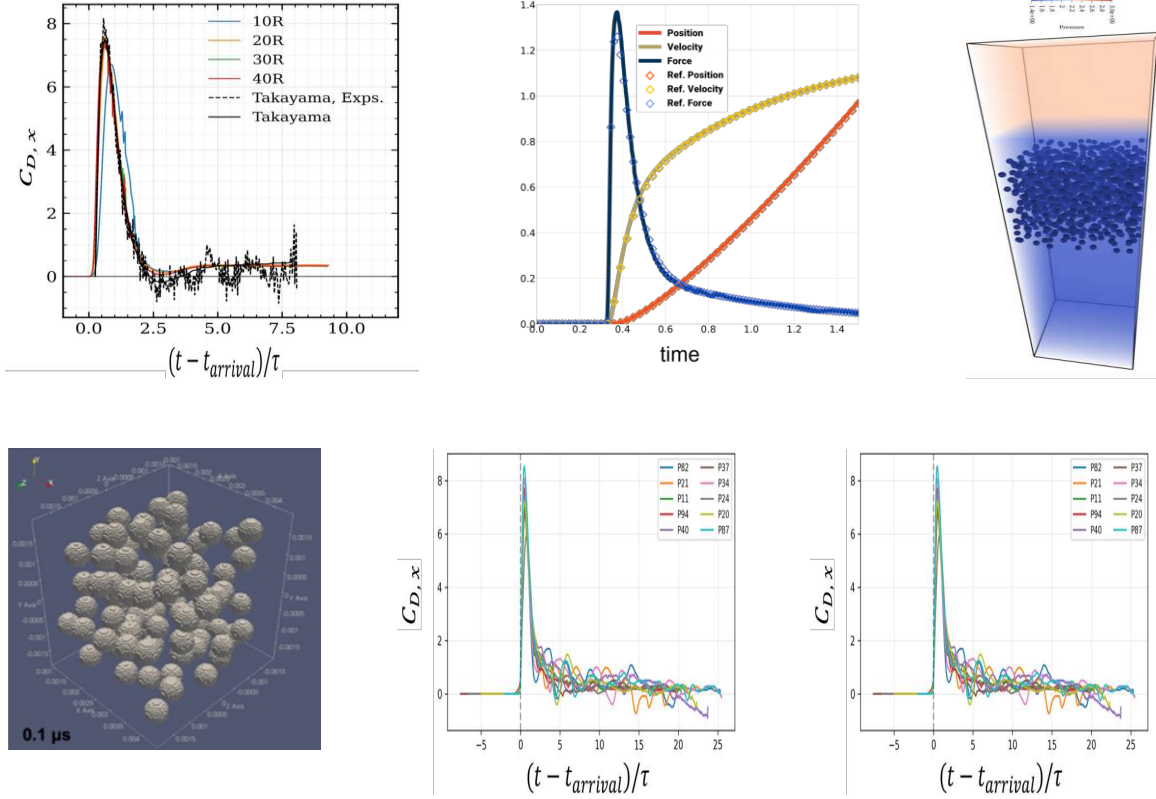


Figure 3: Top row, left to right: (a) W. D., & Schwendeman, D. W. (2006). Moving overlapping grids with adaptive mesh refinement for high-speed reactive and non-reactive flow. *Journal of Computational Physics*, 216(2), 744–779. (b) Case with 1000 IB spherical particle case on only 4 GPUs; grid resolution 800x200x200 and particles diameter is 10 grid cells per diameter; Mach 1.5 incident pressure wave; particles collide with each other and the y/z slip walls. Bottom row, left to right: Our (d) 100 random particle simulation demonstrates that we can handle stochastic arrays, capturing complex (e) drag and (f) torque fluctuations for over 320 million cells in under 2 hours.

2.2.3 EL LES Macroscale Simulations with MFC

At the beginning of the Center, the capabilities of MFC in terms of Euler-Lagrange LES was limited to tracking bubble dynamics. The simulation roadmap requires the ability to track billions of explosively dispersed reactive particles. This required many important additions to the existing code capability; see table 3. With the focus on reactive solid particles, it was found that the existing code structure for bubbles was not sufficient for the Center’s needs. For example, the forces acting on a bubble are very different from the forces for reacting particles. Furthermore, two-way and four-way coupling of reactive particles was considerably different in character. Most importantly, the simulation roadmap requires tracking of billions of reactive particles in a perfectly scalable manner.

Requirement	Previously	Currently
Lagrangian Particle Architecture: Initialization, per-particle arrays, MPI rank ownership, data output	Fully Implemented	Fully Implemented
Fluid-to-Particle Force Application: Interpolation, force models, heat transfer	Not Yet Implemented	Currently Developing
Particle-to-Fluid Field/Force Spreading: Projecting fields and hydrodynamic forces back to Eulerian field	Not Yet Implemented	Currently Developing
Lagrangian Particle Motion: Moving particles, MPI particle transfer, boundary conditions	Not Yet Implemented	Currently Developing
Collision Forces: Inter-particle collisions, wall collisions	Not Yet Implemented	Currently Developing
IB-EL Collision Forces: Resolved particles must be able to collide with sub-grid particles	Not Yet Implemented	Not Yet Implemented
Scaling: MFC with $O(\sim 10^{11})$ moving Lagrangian particles	Not Yet Implemented	Currently Developing
Particle Reactions: Particle mass exchange, heat release, chemistry	Not Yet Implemented	Not Yet Implemented

Table 3: List of tasks accomplished by the macroscale team.

(A) Governing Equations - Lagrangian Particles: The particles are treated as discrete Lagrangian masses, and their time evolution is governed by

$$\frac{d\mathbf{x}_{p,i}}{dt} - \mathbf{u}_{p,i} = 0, \quad m_{p,i} \frac{d\mathbf{u}_{p,i}}{dt} = \mathbf{F}_{gp,i} + \mathbf{F}_{pp,i}, \quad (2.10)$$

$$m_{p,i} C_{p,i} \frac{dT_{p,i}}{dt} = Q_{gp,i}, \quad I_{p,i} \frac{d\boldsymbol{\omega}_{p,i}}{dt} = \boldsymbol{\mathcal{T}}_{gp,i} + \boldsymbol{\mathcal{T}}_{pp,i}, \quad (2.11)$$

where the mass, moment of inertia, position, velocity, and angular velocity of the i^{th} particle are denoted by $m_{p,i}$, $I_{p,i}$, $\mathbf{x}_{p,i}$, $\mathbf{u}_{p,i}$, and $\boldsymbol{\omega}_{p,i}$, respectively. $T_{p,i}$ and $C_{p,i}$ are the i^{th} particle's temperature and specific heat at constant pressure, respectively. In the above equations, $(.)_{gp}$ and $(.)_{pp}$ represent gas-particle and particle-particle interactions, respectively. $(.)_{g,i}$ and $(.)_{p,i}$ denote gas and particle properties evaluated at the i^{th} particle location, respectively. Again, the equations are written such that on the right-hand sides, the phase-coupling closure models of force, heat transfer, and torque are highlighted in red. These closure terms include the hydrodynamic force of the i^{th} particle ($\mathbf{F}_{gp,i}$), the heat transfer from the gas phase to the i^{th} particle ($Q_{gp,i}$), and the hydrodynamic torque on the i^{th} particle ($\boldsymbol{\mathcal{T}}_{gp,i}$). The equations also account for inter-particle interactions, and these terms on the right-hand sides are highlighted in cyan.

Note: Only the models to solve equations 2.10 are currently implemented into MFC's EL solver for solid point particles. The remaining equations 2.11 are currently under construction and will be completed in the coming months.

(B) Particle Projection: Lagrangian particle quantities in the fluid governing equations are converted to Eulerian fields using the projection function $G(\mathbf{x} - \mathbf{X}_i)$, where $\mathbf{X}_i$ is the location of the i^{th} particle. This includes particle volume fraction (ϕ_p) and particle velocity ($\phi_p \mathbf{u}_p$). The particle volume fraction and velocity fields at each cell are approximated as

$$\begin{aligned} \phi_p(\mathbf{x}, t) &\approx \sum_{i=1}^{N_p} G(\mathbf{x} - \mathbf{X}_i) \mathbb{V}_{p,i} \\ \phi_p(\mathbf{x}, t) \mathbf{u}_p(\mathbf{x}, t) &\approx \sum_{i=1}^{N_p} G(\mathbf{x} - \mathbf{X}_i) \mathbf{u}_{p,i} \mathbb{V}_{p,i} \end{aligned} \quad (2.12)$$

where $V_{p,i}$ is the volume of particle i .

The source terms for the fluid must also be projected onto the Eulerian mesh. This includes:

$$S_{mom,x}(\mathbf{x}, t) \approx \sum_{i=1}^{N_p} -G(\mathbf{x} - \mathbf{X}_i) \mathbf{F}_{gp,i,x} \quad (2.13)$$

The projection function decays rapidly with increasing $(\mathbf{x} - \mathbf{X}_i)$ and has compact support. Thus, the momentum and thermal feedback of each particle is localized to the surrounding gas in its immediate neighborhood. Furthermore, G integrates to unity and as a result, momentum and energy conservation between the phases is guaranteed. The sum is over all the N_p particles within the computational domain. Nevertheless, in actual implementation, in evaluating the feedback within each fluid grid cell, one has to perform the summation over only those particles that influence that cell, and therefore the sum over all N_p particles is just a formal representation. In this solver, G is chosen to be a Gaussian filter that spreads the feedback over multiple cells.

The Gaussian kernel function [Balachandar, 2024b] goes as

$$G(\mathbf{x} - \mathbf{X}_i) = \exp \left(-\alpha \left[\left(\frac{x - X_i}{\Delta_x} \right)^2 + \left(\frac{y - Y_i}{\Delta_y} \right)^2 + \left(\frac{z - Z_i}{\Delta_z} \right)^2 \right] \right) \quad (2.14)$$

where α is the Gaussian spreading constant. Here, Δ_x , Δ_y , and Δ_z are the cell sizes of the particle's host cell. We assume the radius of particle i , $r_{p,i}$ to be smaller than the grid size. Note that the Gaussian is not chosen to be spherically symmetric, since we demand that the each particle's effect be spread over multiple cells in each direction. Since spreading is a numerical smoothening construct, there is nothing wrong (or right) about this choice.

For each particle i , the solver spreads the particle's effect across its host cell and M neighboring cells in each direction, where $M = \frac{N-1}{2}$ and N is the total number of cells over which the particles influence is felt in each direction. Here, N can be chosen as $N = 3, 5, 7 \dots$ to determine the number of grid cells over which the Gaussian kernel is discretely evaluated. The Gaussian constant $\alpha = 2.305$ if the spreading is over one cell ($M = 1$) along every direction and $\alpha = 1.023$ if the spreading is over two cells ($M = 2$). With these choices, the Gaussian would have decayed four orders of magnitude at the next cells where it is ignored. The Gaussian function is evaluated at each of the neighboring cells to compute the volume-fraction contribution, hydrodynamic force deposition, heat-transfer sources, and the particle velocity field. To ensure conservation, the total Gaussian kernel sum for each particle is computed. The sum is used to normalize $G(\mathbf{x} - \mathbf{X}_i)$. This sum is represented by

$$G_{\text{sum}} = \sum_{m=-M}^M \sum_{n=-M}^M \sum_{p=-M}^M G(\mathbf{x}_{i+m,j+n,k+p} - \mathbf{X}_i) \Delta x_{i+m} \Delta y_{j+n} \Delta z_{k+p} (1 - \phi_p(\mathbf{x}_{i+m,j+n,k+p})) \quad (2.15)$$

where $\mathbf{x}_{i+m,j+n,k+p}$ is the cell center of the neighboring grid cell relative to the particle host cell. Note that only the volume occupied by the fluid is used for the feedback. The Gaussian kernel is normalized as

$$G^*(\mathbf{x} - \mathbf{X}_i) = \frac{G(\mathbf{x} - \mathbf{X}_i)}{G_{\text{sum}}}. \quad (2.16)$$

Then, projection for volume fraction becomes

$$\phi_p(\mathbf{x}, t) = \sum_i G^*(\mathbf{x} - \mathbf{X}_i) V_{p,i} \quad (2.17)$$

(C) Fluid Interpolation: Fluid quantities at the particle location are computed using high-order Barycentric Lagrange interpolation [Zwick and Balachandar, 2020]. The fluid quantity $A(\mathbf{X})$ at the particle location $\mathbf{X}$ is defined with the fluid quantity at cell i, j, k (a_{ijk}) as

$$A(\mathbf{X}) = \frac{\sum_{k=1}^N \sum_{j=1}^N \sum_{i=1}^N \frac{w_i w_j w_k}{\bar{X}_{ijk}} a_{ijk}}{\sum_{k=1}^N \sum_{j=1}^N \sum_{i=1}^N \frac{w_i w_j w_k}{\bar{X}_{ijk}}} \quad (2.18)$$

where $\bar{X}_{ijk}$ is the product of the distance of the particle to the cell centers in each direction, which goes as $\bar{X}_{ijk} = (\mathbf{X} - x_i)(\mathbf{X} - y_j)(\mathbf{X} - z_k)$. x_i, y_j and z_k are the cell center locations. The weights at each cell in each stencil direction (w_i, w_j and w_k) are computed as

$$w_j = \frac{1}{\prod_{k \neq j} (x_j - x_k)}; \quad j, k = 1, \dots, N \quad (2.19)$$

The weights depend only on the local grid spacing in each direction for the stencil. Thus, the weights for all possible stencils are precomputed at the simulation start time for efficiency. This method accounts for variable grid spacing in any direction and various orders of accuracy. 2nd-order accuracy is achieved with a 3-point stencil centered around the cell containing the particle, with $N=3$. 4th order accuracy corresponds to a 5-point stencil around the particle's cell, with $N=5$.

Note: Both the projection and interpolation algorithms have been successfully integrated into MFC. Verification tests were carried out to show correctness for a variety of conditions.

(D) Force Coupling Models: The hydrodynamic forces presently take into account the following force contributions:

$$\mathbf{F}_{gp,i} = \mathbf{F}_{qs,i} + \mathbf{F}_{pg,i} + \mathbf{F}_{am,i} \quad (2.20)$$

The contributions on the right-hand side, in order, are the average quasi-steady force, the undisturbed flow force (also known as the pressure gradient force), and the average added mass force. Each of the contributions accounts for an underlying momentum coupling mechanism in the following way:

- The quasi-steady force depends on the instantaneous state of the particle relative to the surrounding fluid. In the compressible regime, the quasi-steady force depends on the mesoscale state of flow around the particle characterized in terms of the following parameters: particle Reynolds number based on relative velocity ($Re_{p,i}$), the Mach number based on relative velocity ($M_{p,i}$), the local volume fraction of particles ($\alpha_{p,i}$), and the microscale state of the neighboring particles.
- The pressure gradient force is the undisturbed flow force that exists even in the absence of the perturbing presence of the i^{th} particle.
- The added mass force depends on the relative acceleration of the particle with respect to the surrounding fluid, denoted as $([D\mathbf{u}_g/Dt]_i - d\mathbf{u}_{p,i}/dt)$, where subscript i denotes evaluated at the particle. Due to its inviscid origin, the added mass force is independent of particle Reynolds number, but depends on $M_{p,i}$ and volume fraction [Parmar et al., 2008, Briney et al., 2025]. The presence of neighbors affects added mass in two ways. First, the added mass force on the i^{th} particle that depends on its acceleration with respect to the surrounding fluid is influenced by the presence of the nearby particles. Second, there is an additional induced mass force due to the acceleration of the neighbors.

The most recent and best available model is used for each of these force contributions. The sources of these most recent models are presented in Table 4. The third column of the table also lists the key input parameters required for the closure model. Finally, in the comment column, we provide additional information documenting the alterations that needed to be made to make the model suitable for practical EL implementation.

Table 4: Details of all the phase-coupling closure models of force, heat transfer, and work implemented in MFC. The primary sources of these models, along with the key variables that serve as their inputs, are also presented. Several of these models required subtle modifications to make them suitable for implementation in an EL simulation. These modifications are summarized under “Comments” in the last column of the table.

Model	Source	Input	Comments
Quasi-steady average	Loth <i>et al.</i> [Loth <i>et al.</i> , 2021] Osnes <i>et al.</i> [Osnes <i>et al.</i> , 2023] Parmar <i>et al.</i> [Parmar <i>et al.</i> , 2010] Gidaspow [Gidaspow, 2004]	$\text{Re}_{p,i}, M_{p,i}, \phi_{p,i}, C_{D,i}$.	Enforced continuity across regimes & avoided singularity in Loth model.
Added mass average	Béguin <i>et al.</i> [Béguin <i>et al.</i> , 2016] Parmar <i>et al.</i> [Parmar <i>et al.</i> , 2008]	$\text{V}_{p,i}, C_M, \rho_{g,i}, \mathbf{u}_{g,i}, \mathbf{u}_{p,i}$.	Béguin model for $\phi_{p,i}$ correction. Parmar model for $M_{p,i}$ correction.
Collision	Balachandar [Balachandar, 2024b]	$\mathbf{x}_{p,i}, \mathbf{x}_{p,j}, \mathbf{u}_{p,i}, \mathbf{u}_{p,j},$ $\boldsymbol{\omega}_{p,i}, \boldsymbol{\omega}_{p,j}$.	Normal and tangential components included.

Note: The force coupling models of quasi-steady, pressure gradient, and added mass have all been incorporated into MFC and tested. Additional force models will be included in the coming months.

(E) Collisional Force: The collisional force between the i^{th} particle and its j^{th} neighbor is modeled as soft sphere collision. Allowing for the simultaneous contact between the i^{th} particle and N_c of its neighbors, we obtain

$$\mathbf{F}_{pp,i} = \sum_{\substack{j=1 \\ j \neq i}}^{N_c} (\mathbf{F}_{pp,ij}^n + \mathbf{F}_{pp,ij}^t), \quad (2.21)$$

where the normal and tangential forces are given respectively by

$$\mathbf{F}_{pp,ij}^n = -\kappa_n \delta_{ij} \mathbf{n}_{ij} - \eta_n \mathbf{u}_{n,ij} \quad (2.22)$$

and

$$\mathbf{F}_{pp,ij}^t = -[\min(\mu_c |\mathbf{F}_{pp,ij}^n|, \eta_t |\mathbf{u}_{t,ij}|)] \mathbf{t}_{ij}, \quad (2.23)$$

where $\delta_{ij} = (r_{p,i} + r_{p,j}) - |\mathbf{x}_{p,j} - \mathbf{x}_{p,i}|$ is the overlap distance between the colliding objects, $\mathbf{n}_{ij}$ the unit vector from the i^{th} particle to the j^{th} particle, μ_c the coefficient of Coulomb friction, and η_t the damping coefficient. The relative velocity between the particles i and j is given by

$$\mathbf{u}_{p,ij} = \mathbf{u}_{p,j} - \mathbf{u}_{p,i} + (r_{p,j} \boldsymbol{\omega}_{p,j} + r_{p,i} \boldsymbol{\omega}_{p,i}) \times \mathbf{n}_{ij}, \quad (2.24)$$

where $r_{p,i}$ and $r_{p,j}$ are the particle radii, and $\boldsymbol{\omega}_{p,i}$ and $\boldsymbol{\omega}_{p,j}$ are the corresponding angular velocities. The relative velocity can be decomposed into a normal and tangential component via

$$\mathbf{u}_{n,ij} = (\mathbf{u}_{p,ij} \cdot \mathbf{n}_{ij}) \mathbf{n}_{ij}, \quad \mathbf{u}_{t,ij} = \mathbf{u}_{p,ij} - \mathbf{u}_{n,ij}. \quad (2.25)$$

The tangent unit vector that points in the direction of $\mathbf{u}_{t,ij}$ is $\mathbf{t}_{ij} = \mathbf{u}_{t,ij}/|\mathbf{u}_{t,ij}|$.

To compute the normal collisional component, the spring stiffness coefficient κ_n is chosen as the minimum of two values: (i) a dynamic model that depends on the integration time-step Δt , spring

material stiffness (k_{sp}), and particle mass ($m_{p,i}$), and (ii) a physical model that depends on the Poisson's ratio (ν_k) and Young's modulus (E_k) of the colliding particles [Balachandar, 2024b]

$$\kappa_n = \min \left(\frac{\pi^2 m_{p,i}}{(\Delta t k_{sp})^2}, \frac{4}{3} E^* \sqrt{R^*} \sqrt{|\delta_{ij}|} \right), \quad (2.26)$$

with,

$$E^* = \left(\frac{1 - \nu_1^2}{E_1} + \frac{1 - \nu_2^2}{E_2} \right)^{-1}, \quad R^* = \frac{r_{p,i} r_{p,j}}{r_{p,i} + r_{p,j}}. \quad (2.27)$$

The normal force damping coefficient is given in terms of the coefficient of restitution ϵ as

$$\eta_n = - \frac{2 \sqrt{\kappa_n} \ln \epsilon}{\sqrt{\ln^2 \epsilon + \pi^2}} \left(\frac{1}{m_{p,i}} + \frac{1}{m_{p,j}} \right)^{-1/2}. \quad (2.28)$$

Note: Only the normal collisional force model has been incorporated into MFC. The tangential collisional model will be included in the coming months.

(F) EL Results: Selected results highlighting the rapid progress of the Center of Euler-Lagrange capabilities implemented into MFC by the macroscale team. Figure 4 shows the performance and capability of MFC for the hemi-spherical blast problem.

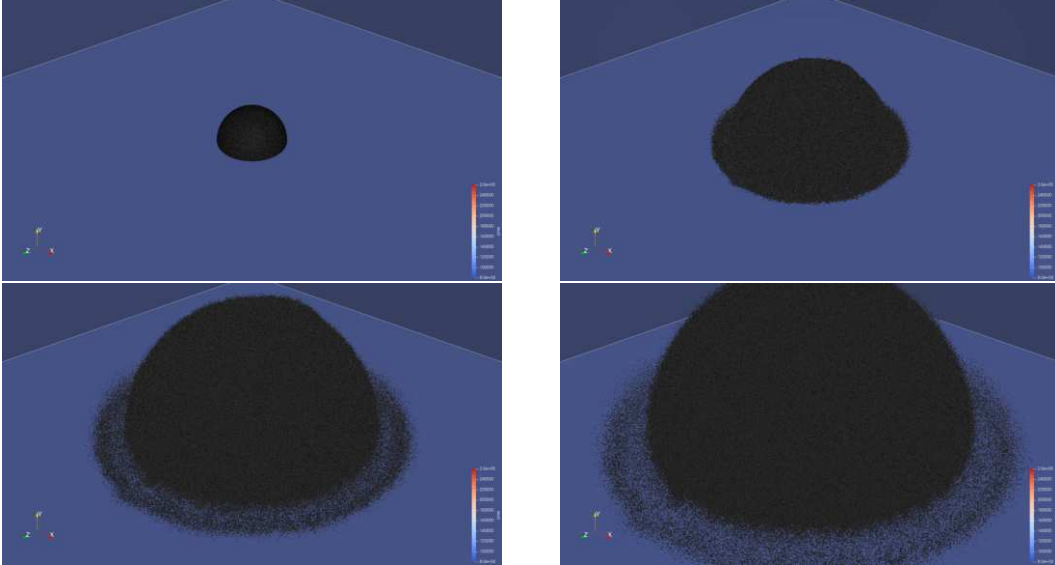


Figure 4: Simulation of hemispherical explosive surrounded by a bed of particles using the recent updates to the EL solver in MFC. Four time instances are shown.

2.2.4 MFC Development and Exascale Performance

The Multicomponent Flow Code (MFC) is an exascale accelerator-performant CFD solver [Bryngelson et al., 2021] and 2025 Gordon Bell Finalist [Wilfong et al., 2025b] with demonstrated near-roofline performance on CSCS Alps (GH200), OLCF Frontier (Trento+MI250X), and LLNL El Capitan (MI300A). Recently, MFC has earned recognition as a 2025 Gordon Bell Prize Finalist.

MFC is currently being used on LLNL Toulumne for the PSAAP IV effort. MFC will be applied to a wide range of scales and physics relevant to the demonstration problem of group combustion.

The different simulations require Euler–Lagrange (EL) and Euler–Euler (EE) simulations of the demonstration problem of explosive dispersal of burning particles, which will employ the best-possible machine learning (ML)-based closure models developed using particle-resolved DNS (PRDNS) microscale-informed understanding of group combustion (see fig. 5).

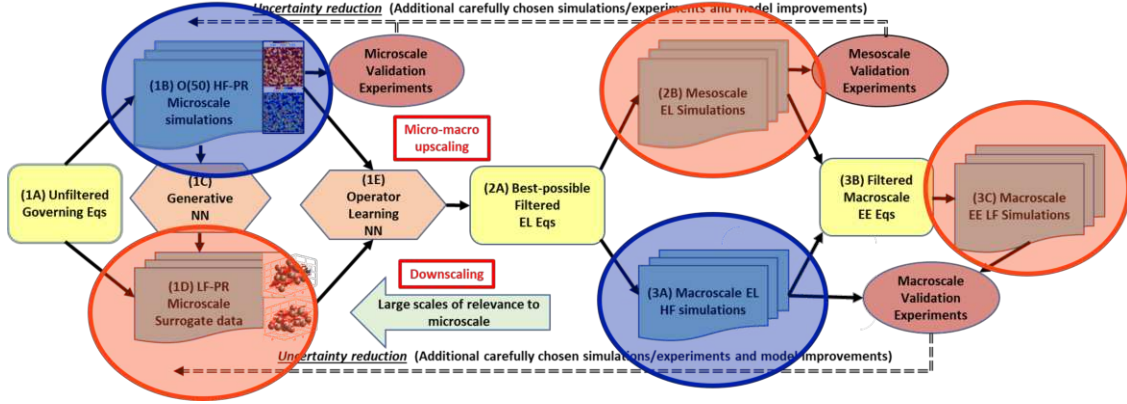


Figure 5: MFC will be used to study group combustion at micro, meso, and macroscales. The different scales will require exascale computations using particle-resolved (PR), Euler–Lagrange (EL), and Euler–Euler (EE) methods.

Numerous capabilities have been integrated into MFC over the past six months; see table 2 and table 3. These capabilities were added to MFC to capture state-of-the-art multiphase physics and chemistry models while maintaining exascale performance. In particular, these capabilities will enable MFC to conduct exascale simulations across the micro, meso, and macro scales. For details about the capabilities of MFC as of 2026, please read the following papers [Bryngelson et al., 2021, Schmidmayer et al., 2020, Radhakrishnan et al., 2024, Wilfong et al., 2025a,b, 2026].

Here, we list the new features needed in MFC to simulate the problems in group combustion considered by CM³C. Completed features are noted. Details of the physical models are described in the other sections.

(A) PR DNS: The immersed boundary (IB) method for resolving the flow details of finite-sized moving particles for particle-resolved (PR) simulations is currently being implemented into MFC by the microscale team.

1. Periodicity: The particle-resolved simulations require periodic boundary conditions in one, two, or three directions. Here, periodicity applies to both the fluid and the particles. For the fluid phase, implementing the periodic boundary condition was straightforward. In the case of resolved particles, the periodic boundary condition means parts of a particle may be split across the domain boundaries. As a particle boundary crosses a face, edge, or corner of the computational domain, parts of the particle are split across periodic boundaries and must be carefully handled in a multi-core calculation, since they can be assigned to different cores. When moving particles, how they are partitioned across processors must be carefully handled.
2. Performance improvements: IB compute is now fully supported on the GPU. Processing the mesh into memory, IB marker generation, and levelset computation have been parallelized. Interpolation of IB markers was removed in favor of a projection-based distance search. This means that we no longer need to interpolate (saving significant per-processing time and memory), we iterate over significantly fewer vertices (by a factor of 50 on the coarsest grids),

and we obtain exact levelset values rather than approximations based on the interpolated mesh’s resolution. Unified distance and normal compute. We no longer perform a distance search and then a separate normal vector search. This is all handled in a single subroutine, cutting search time in half.

3. Improvements to STL/arbitrary geometries: Modified STL generation to be cell center-based instead of volume fraction-based. This prevents edge cases in which the cell center is outside the model, but the point is labeled as inside. This error leads to level sets that point into the immersed boundaries, resulting in numerical instability.
4. Inter-particle collisions: Collision algorithm required for particle–particle and particle–wall interaction. This feature has now been implemented using a soft sphere collision algorithm.
5. IO of particle quantities: All the particle-related characteristics at each time step for each MPI rank must be computed as a function of particle index i and time t . This includes the following long list of quantities needed for the downstream machine learning pipeline:

$$(i, t, d_p, x_p, y_p, z_p, u_p, v_p, w_p, a_x, a_y, a_z, T_p, F_x, F_y, F_z, \mathcal{T}_x, \mathcal{T}_y, \mathcal{T}_z)$$

6. IO of filtered quantities: Compute the “filtered quantities” that are needed for subgrid stress and all other closure quantities at user-specified frequency. For example, for subgrid stress, we need to compute and write both $\overline{u_i u_j}$ and $\overline{u_i}$. These raw filtered quantities can be post-processed to calculate the subgrid stress as $\overline{u_i u_j} - \overline{u_i} \overline{u_j}$. The key is to use the smallest possible filter size in MFC, which will be of the order of the particle size. These filtered quantities should be written on a coarser mesh than that used for PRDNS (to be determined by the filter size) and will thus be orders of magnitude smaller. This strategy is optimal in many ways. ML modeling requires closure terms to be computed for varying filter sizes, but computing all the different filter calculations in MFC is suboptimal. The advantage of using the smallest filter size in MFC is that this data can be filtered again with larger filters. This way, we can compute the closure quantities for multiple filter sizes as a post-processing step.

(B) Euler–Lagrange (EL) methods: The macroscale team is currently incorporating needed algorithms and physics into MFC for the meso and macroscale simulation problems.

1. MPI/multi-rank distributed memory, two-way coupled capability: This effort adds support for moving Euler–Lagrange particles and various force models. Tracer particles that simply follow the background flow are implemented, as are Newton’s 2nd Law particles that respond to body forces, pressure gradients, and drag.
2. EL particle tracking: Developed a set of subroutines for particulates, separate from the existing EL bubble routines.
3. Interpolation: Extended the existing capabilities to include second and fourth order interpolation of fluid quantities to the particle centroids using an efficient barycentric algorithm [Zwick and Balachandar, 2020].
4. Inter-particle EL collisions: Integrated collision detection and soft sphere collision algorithms within the current MFC EL framework.
5. Force modeling: Included most of the state-of-the-art compressible gas-particle coupling force and feedback terms that are in ppiclf to MFC [Daoud et al., 2026].
6. IO: Similar output of particle characteristics as in PRDNS.

7. Capability: Incorporation of ML models directly into MFC. In this case, the force calculations performed by MFC or ppiclf will become an optional capability.

2.2.5 Experiments

The experimental team, in collaboration with the micro and macroscale simulation teams and the UQ team, have made great progress toward the planning and characterization of the various components of the carbon rod experiment (for validation of chemistry and several aspects of group combustion modeling) and the hemispherical blast problem (for validation of instabilities, flame location, etc, at the larger spatial scales). For planning details see table 5.

Task	Year 1	Year 2	Year 3	Year 4	Year 5
Explosive demonstration using standard imaging and pressure gauges			▲		
Single rod experiments validated against published literature		▲			
Design/build upgraded multi-rod configurations					
Multi-rod group combustion measurements			▲		
Sub-scale testing of blast-hardened LAS/emission/extinction gauges					
Design/build multi-rod configurations with spacing guided by UQ team					
Multi-rod, multi-parameter experimental measurements guided by UQ team					▲
Demonstration experiments with LAS and other advanced diagnostics				▲	
Final Publication/documentation					▲

Table 5: CM³C Experimental Schedule.

(A) Carbon Rod Experiments: Microscale experiments will investigate the combustion of closely-spaced, mm-sized carbon rods within a high-speed airflow (fig. 2 (a)). Flow velocities up to Mach 1.5 and stagnation temperatures of 500 K will mimic some of the most extreme relative air conditions experienced by the fastest moving particles in the explosive configuration. Individual carbon rods will be placed within this flow and Joule-heated to control surface temperatures. Videography will quantify the time-resolved changes in the surface profiles and overall reaction rates. This configuration is inspired by experimental methodologies, which are well-established for low-speed, isolated particle configurations [Makino et al., 1994, 1998, 2003, Makino and Umehara, 2007] where results have already been proven useful for validation of HR-PR simulations [Ou et al., 2022, Karchniwy et al., 2022, Jiang et al., 2022, Luo et al., 2018, Gao et al., 2014].

Figure 6 (left) shows the in-progress experimental configuration where labels identify the components of the setup. In the image shown in the lower left panel, bright emissions from the Joule-heated carbon rod are clearly visible. Preliminary measurements of the surface regression as a function of surface temperature are shown in fig. 6 (right) with points colored by axial coordinate along the rod. Results in fig. 6 (right) quantify increasing surface regression rates with increasing temperature, in agreement with prior literature [Makino et al., 1994, 1998, 2003, Makino and Umehara, 2007]. Such data are invaluable to the scientific community at large, and will help validate the Center’s reactive flow models used at the microscale. The Center’s upcoming work will quantify measurement uncertainty, integrate high-speed air flows, and progress to novel multi-rod configurations.

(B) Hemispherical Blast Experiments: Figure 7 (upper left) shows the schematic of our demonstration problem consisting of a high explosive (HE) hemispherical charge surrounded by reactive carbon (C) or magnesium (Mg) particles. Detonation of the HE rapidly shocks, heats, and accelerates the particles. As the particulate front radially expands, it undergoes complex instabilities and reactions [Frost et al., 2002, 2012, Frost, 2018, Posey et al., 2021]. Molten Mg burns within vapor diffusion flames [Dreizin, 2000, Legrand et al., 2001, Dreizin, 2014] while carbon undergoes

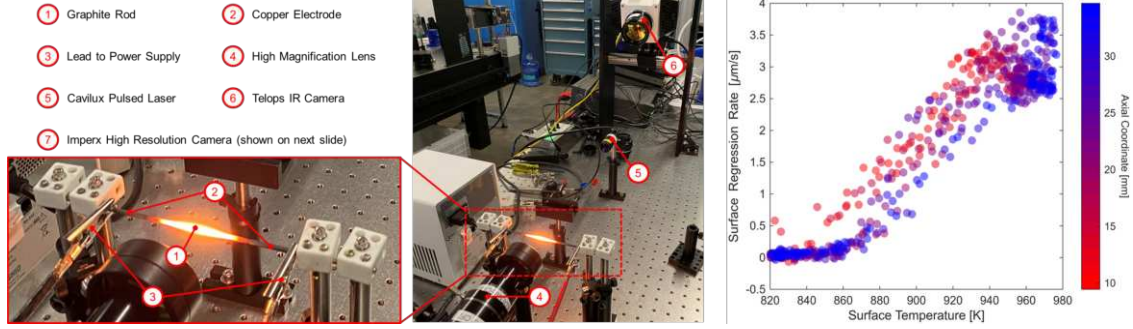


Figure 6: Left: Current experimental setup. Right: Surface regression as a function of surface temperature.

heterogeneous surface and gas-phase reactions [Willhardt et al., 2024, Michelsen, 2017, Bradley et al., 1985]. Thus, C and Mg are selected to capture these two ends of particle burning regimes. Predicting the reactive particle-gas and particle-gas-particle interactions across the broad range of scales within the expanding post-detonation flow poses an immense computational challenge. To meet this challenge, the experimental team have been planning the experiments for the demonstration problem, with input from the macroscale team. The setup is shown in fig. 7 (lower left); the planned diagnostics in fig. 7 (top right); and the preliminary results for a 12-mm diameter PETN hemisphere coated with SiO_2 particles are shown in fig. 7 (bottom). The Center's upcoming work will perform similar measurements for Mg and C coated hemispheres using the diagnostics illustrated in the upper right panel of fig. 7. In addition, interface-resolved simulations of the carbon rod experiments will begin during the next six months.

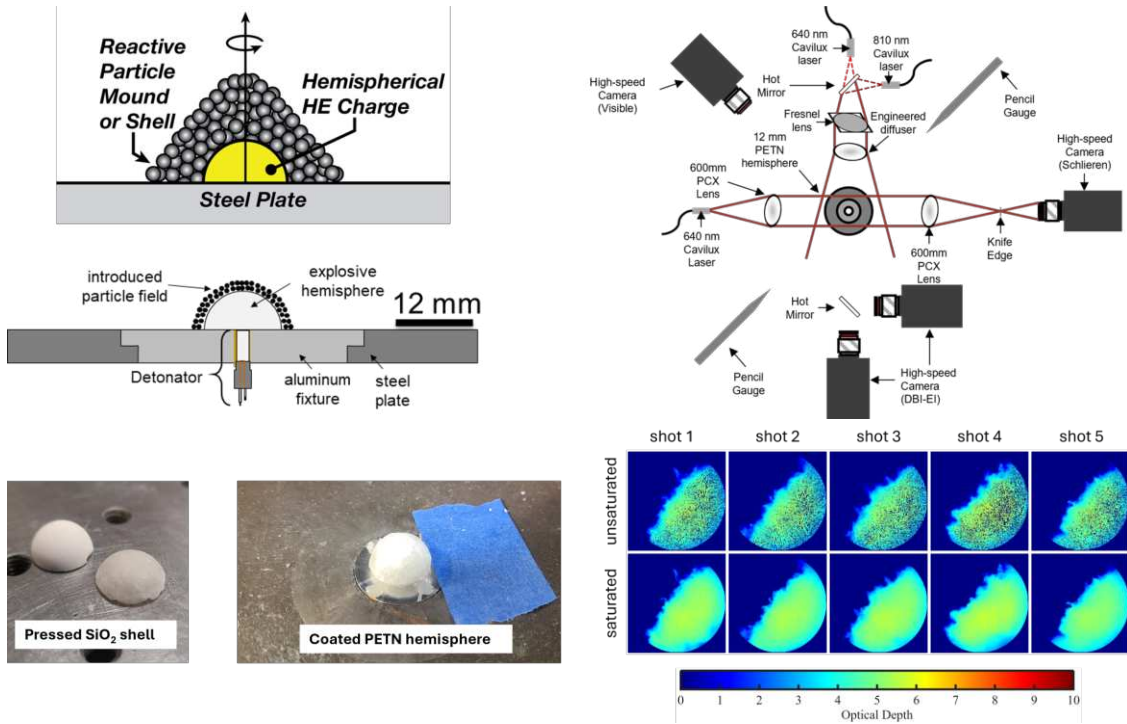


Figure 7: The configuration for the hemispherical blast problem (left), the measurement configuration (upper-right), and preliminary measures of extinction due to a 12-mm SiO_2 coated PETN hemisphere.

(C) **Companion Simulations:** The macroscale team has been carrying out companion simulations

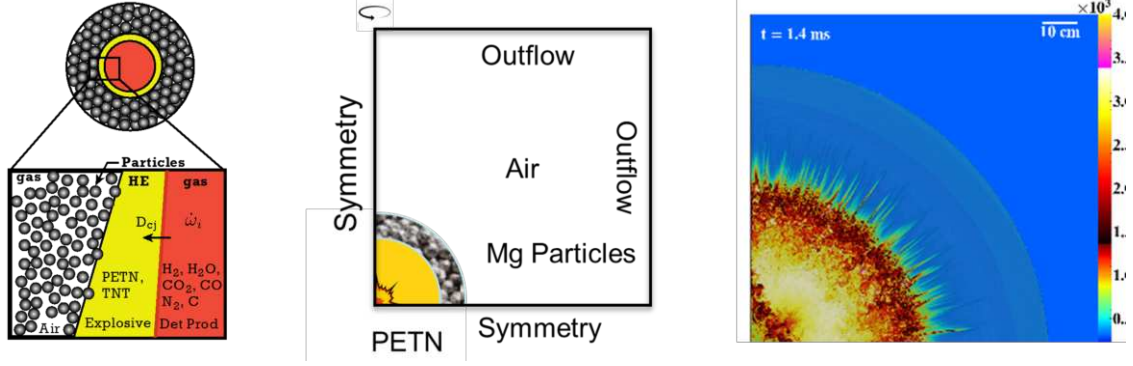


Figure 8: Left: Computational figure of the explosive and particle bed. Middle: Computational domain, Right: Snapshot showing gas-phase temperature.

of the demonstration problem using the advanced CFD solver, HyBurn, [Houim and Oran, 2016, Houim, 2021]. The code is based on the following: (i) A diffuse interface method (DIM) is used for reactive gas and high-explosive (HE); (ii) a novel detonation burn model converts HE to products at the CJ composition; (iii) an Eulerian multiphase model is used for the particles; (iv) the code accounts for phase change, drag, heat transfer, collisions, and compaction; and (v) particles burn via an empirical burning model turbulence modeled using ILES. The code uses similar explosives as the experimental team, namely PETN. The hemispherical explosive and particle characteristics are representative of the expected conditions proposed by the experimental team.

The goal of this work is primarily to guide the experimental team during the design stage, providing expected shock/blast location as a function of time for probe sensor locations, expected blast radius, flame front location, etc. The model uses an empirical burning time correlation for the magnesium particles [Lim et al., 2014]. Ignition is modeled using an Arrhenius-like ignition delay model [Kuhl and Balakrishnan, 2012, Medvedev et al., 1982]. Figure 8 shows the computational setup and a preliminary result on the ignition and combustion of PETN-dispersed magnesium particles. Upcoming work will be to analyze these results and begin preliminary comparisons to experimental measurements.

2.2.6 Machine Learning Model Development

Developing efficient solution strategies that accurately predict large N -particle systems without resorting to expensive computations is an important goal. Towards our overarching goal of developing ML models for group combustion of the explosive dispersal problem, we start with the simpler N -particle problem, in which N particles are randomly distributed and settle in an otherwise quiescent fluid. The entire fluid motion is driven by the moving particles. The ML pipelines set up for the simpler problem will be repurposed to solve the group combustion problem. The HF-PR data needed for training group combustion ML closure models will not be available till year two. Therefore, the ML model development effort has begun with the simpler problem for which we have the required data.

At some instance in time, let the i^{th} particle be of diameter d_i , located at $\mathbf{x}_i$, and let the translation and rotation velocities be $\mathbf{v}_i$ and $\mathbf{\Omega}_i$. We consider three N -particle prediction questions. These are precisely the same prediction questions that we will face with the group combustion problem as well.

- **Generative Flow-Prediction:** Given the particle information, and given an inference point $\mathbf{y}$, we want to predict the velocity $\mathbf{u}$ and pressure p at that inference point. With this prediction capability, we can choose $\mathbf{y}$ to be anywhere (including inside the particles), and thus we get the ability to predict the velocity and pressure fields (i.e., $\mathbf{u}(\mathbf{y})$ and $p(\mathbf{y})$) for the N -particle problem.
- **Force-Prediction:** We note that the above flow prediction is needed only to calculate the force and torque on the N -particles. So, what if we calculate the force and torque directly? Thus, given the particle information, in this prediction quest, we want to predict the force $\mathbf{F}_i$ and torque $\mathbf{T}_i$ of all the $i = 1, 2, \dots, N$ particles.
- **Subgrid-Prediction:** Note that in addition, we also need to predict the subgrid quantities. This prediction is somewhere between flow and force prediction. Subgrid quantities are field variables, but they are far smoother. Their prediction task can be stated as follows: Given particle information and an inference point $\mathbf{y}$, predict the subgrid quantity $\mathbf{R}$ at that point. There are several subgrid quantities; we just use $\mathbf{R}$ as an example.

Several additional comments are needed. First, only “Generative Flow-Prediction” can be from first principles. I.e., we know the fundamental equations that govern $\mathbf{u}(\mathbf{y})$ and $p(\mathbf{y})$. Therefore, for flow-prediction we will use both data-driven and physics-informed neural network (PINNs) approaches. This is why we use the term “generative” for flow-prediction. For the force-prediction and subgrid-prediction tasks, we will only use the data-driven approach.

Second, let us consider the typical amount of data available for data-driven approaches. For flow-prediction, $\mathbf{u}(\mathbf{y})$ and $p(\mathbf{y})$ Data is typically available on a very fine grid. For force-prediction, force $\mathbf{F}_i$ and torque $\mathbf{T}_i$ information are only one per particle (thus far fewer). For subgrid-prediction, $\mathbf{R}$ information is typically available on a coarse mesh. Its data availability will be two orders of magnitude lower than for flow-prediction, but still much higher than for force-prediction. In the following sections, we will address our plan and execution of the different prediction strategies.

Generative Flow-Prediction using PINNs:

We start with the following proposition: At some instance in time, if there are N particles of diameter d_i , located at $\mathbf{x}_i$, translating and rotating with velocities $\mathbf{v}_i$ and $\mathbf{\Omega}_i$, they induce a quasi-steady flow of velocity $\mathbf{u}$ and pressure p . This quasi-steady flow satisfies the steady Navier-Stokes equations, with velocity satisfying no-slip and no-penetration on the surface of the particles¹, which are symbolically represented as

$$\begin{aligned}
\text{fluid domain:} \quad & \mathcal{N}(\mathbf{u}(\mathbf{y}), p(\mathbf{y})) = 0 \quad \text{for } \mathbf{y} \in V_f \\
\text{far field:} \quad & \mathbf{u}|_{\mathbf{y} \rightarrow \infty} = 0 \\
\text{on the particle:} \quad & \mathbf{u}|_{\mathbf{y} \in B_i} = \mathbf{v}_i + \mathbf{\Omega}_i \times \mathbf{n} \quad \text{for } i = 1, \dots, N,
\end{aligned} \tag{2.29}$$

where V_f is the fluid domain exterior of all the particles, where the Navier-Stokes equation applies. The surface of the i^{th} particle is given by $B_i := |\mathbf{y} - \mathbf{x}_i| = d_i/2$, and $\mathbf{n}$ is the surface normal.

The above N -particle problem is fully coupled in the sense that the flow around all the particles must be solved together. It is not the sum of the flow around N isolated particles. Let us elaborate on this further. Let us define $\mathbf{u}_i$ and p_i as the flow around the i^{th} particle, in the absence of all

¹The actual flow is unsteady, satisfies the unsteady Navier-stokes equations, and will also depend on the instantaneous acceleration of the particles. With the quasi-steady assumption, we ignore this unsteady nature of the flow.

other particles, given by

$$\begin{aligned}
\text{fluid domain:} \quad & \mathcal{N}(\mathbf{u}_i(\mathbf{y}), p_i(\mathbf{y})) = 0 \quad \text{for } \mathbf{y} \in V_{f,i} \\
\text{far field:} \quad & \mathbf{u}_i|_{\mathbf{y} \rightarrow \infty} = 0 \\
\text{on the particle:} \quad & \mathbf{u}_i|_{\mathbf{y} \in B_i} = \mathbf{v}_i + \boldsymbol{\Omega}_i \times \mathbf{n},
\end{aligned} \tag{2.30}$$

where $V_{f,i}$ is the fluid domain exterior of only the i^{th} particle. The important difference is that the last condition now applies only to the i^{th} particle, whereas in the coupled problem it applied over all the particles.

Superposition We first explain the simple-minded approach of summing up the isolated solutions of all the particles. The resulting naive superposition can be expressed as

$$\mathbf{u}^{\text{naive}}(\mathbf{y}) = \sum_{j=1}^N \mathbf{u}_j(\mathbf{y}) \quad \text{and} \quad p(\mathbf{y}) = \sum_{j=1}^N p_j(\mathbf{y}). \tag{2.31}$$

In other words, we have approximated the flow generated by all the N particles to be the simple sum of the flow due to each particle. This is a good approximation if the particles are far apart and do not interact. It is easy to see the problem if the particles approach each other. The fluid velocity at the surface of the i^{th} particle is given by

$$\mathbf{u}^{\text{super}}(\mathbf{y} \in B_i) = \sum_{j=1}^N \mathbf{u}_j(\mathbf{y} \in B_i) = \underbrace{\mathbf{v}_i + \boldsymbol{\Omega}_i \times \mathbf{n}}_{\text{self effect}} + \underbrace{\sum_{j=1, \neq i}^N \mathbf{u}_j(\mathbf{y} \in B_i)}_{\text{neighbor effect}}. \tag{2.32}$$

As can be seen, while the self-effect correctly gives the correct fluid velocity at the surface of the i^{th} particle, the induced velocity due to all other particles, represented as the “neighbor effect”, will in general be non-zero. Thus, the velocity field resulting from naive superposition will not satisfy the velocity boundary condition on any of the particles.

The final superposed solution can be obtained as for any inference point $\mathbf{y}$. The effect of each particle on the velocity and pressure at the inference point can be calculated using the procedure outlined in §(2.3.1) by using the correct modified velocities of each particle. The accuracy of implicit superposition can be evaluated by comparing to the true velocity and pressure obtained by solving (2.29) for the coupled problem. We calculate the pointwise error as

$$error_u(\mathbf{y}) = |\mathbf{u}^{\text{super}}(\mathbf{y}) - \mathbf{u}(\mathbf{y})|^2 \quad \text{and} \quad error_p(\mathbf{y}) = |p^{\text{super}}(\mathbf{y}) - p(\mathbf{y})|^2 \tag{2.33}$$

These pointwise errors can be averaged over the fluid volume V_f to obtain MSE. Note that we have the ground truth information or the solution of the coupled problem.

Generative Data-Driven Flow-Prediction:

The generative data-driven problem fits nearly the same description as the initial proposition given for generative flow-prediction using PINNs. That is, we wish to predict the velocity $\mathbf{u}$ and pressure p of quasi-steady flow induced by N particles of diameter d_i , located at $\mathbf{x}_i$, translating with velocities $\mathbf{v}_i$. The major difference at this time is the assumption of no rotation of the particles. Our initial approach to data-driven flow prediction involves training a structured neural network to predict fluid velocity $\mathbf{u}(\mathbf{y})$ induced at any inference point $\mathbf{y}$ by a single particle, then superposing the results

as described above. This neural network has multiple branches which process separate inputs to enforce specific expected behaviors.

The first branch of the neural network consists of standard feedforward layers and processes exclusively scalar data. This branch primarily adapts to the greater context of the particle’s environment. It receives inputs of a single particle’s Reynold’s number Re , particle density ratio PDR, and local particle volume fraction ϕ . It outputs n scalars, which can be interpreted as the coefficients of an empirical vector field decomposition.

The second branch of the neural network consists of rotationally-equivariant e3nn Geiger and Smidt [2022] layers and processes exclusively vector data. This ensures that rotations and reflections about the particle produce consistent results and reduces the required amount of training data. This branch receives inputs of the relative position of the inference point to the particle $\mathbf{y}_i$ and a unit vector indicating the direction of the particle’s motion $\mathbf{e}_i$. It outputs n vectors, which can be interpreted as the modes of an empirical vector field decomposition.

The third branch of the neural network consists primarily of mathematical operations which are structured to apply the learned weights of a feedforward neural network as coefficients and offset of a pseudo-ellipsoid which surrounds the particle and extends primarily along the axis of its motion. A sigmoid function is applied to the initial scalar output. Thus in the interior of the ellipsoid, this branch produces an output of approximately 1, while on the exterior, it produces an output of approximately 0, with a gradual transition at the boundary. This can be interpreted as a distance-based decay factor.

The coefficients and modes, produced by the first and second branches respectively, are multiplied element-wise and linearly combined into a single vector output as an initial estimate of the flow velocity at the inference point, $\mathbf{u}'(y)$. The third branch’s output is then multiplied by the initial estimate to produce a final estimate, $\mathbf{u}(y)$.

We train this neural network on the non-dimensionalized results of particle-resolved DNS (PR-DNS). In particular, our initial training uses one time step of a simulation involving 64 identical particles with triply periodic boundary conditions. The training loss is the mean-squared error (MSE) between PR-DNS results and the sum across particles (naïve superposition) of predicted velocity at the inference points. Validation is performed using samples from a second time step to optimize network hyperparameters such as n , the number of outputs for branches one and two, and r_{train} , the radius around each particle from which training data is taken. Final test results are given based on samples from a third time step.

With our best-case hyperparameters, over the entire simulation, the MSE in the non-dimensionalized velocity is 2.987 (compared to 12.109 if guessing the mean velocity of the simulation), and we achieve an R^2 -score of 0.753. If we restrict our evaluation to the regions within 1 particle diameter of a particle center, the MSE in the non-dimensionalized velocity is 1.875 (compared to 34.386 if guessing the mean velocity of the simulation), and we achieve an R^2 -score of 0.909. Preliminary work on an equivalent network for pressure $p(\mathbf{y})$ prediction indicates similar performance can be expected.

2.2.7 Group Combustion

Droplet combustion is a problem of great importance in many industrial applications. The design and operation of diesel engines, power plants, and jet engines, to name just a few, depend on our understanding of how fuel droplets burn in an oxidizer environment. The vaporization and burning

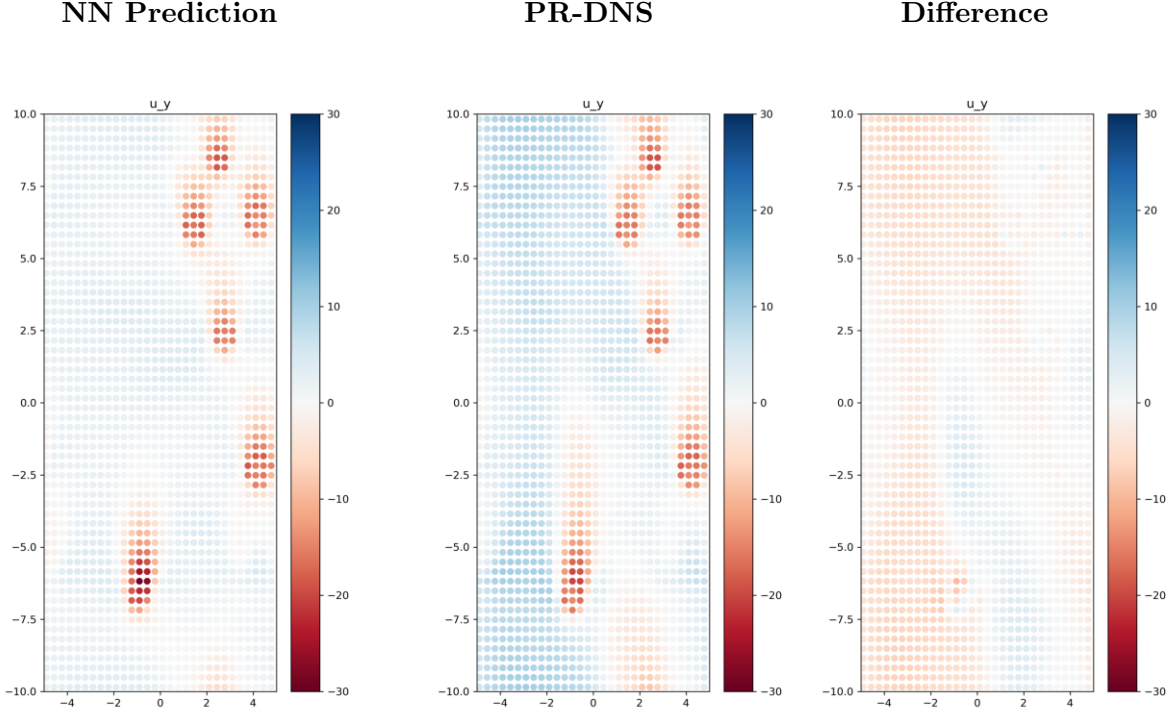


Figure 9: Vertical component of flow velocity for test data inference points.

of liquid fuel droplets form the fundamental building block of spray combustion encountered in propulsion systems, gas turbines, diesel engines, furnaces, and accidental fire scenarios. As a result, the fundamental study of droplet combustion as a scientific topic in fluid mechanics has attracted great attention [Faeth, 1979, Law, 1982, Sirignano, 1983, Williams, 2018].

The earliest investigations focused on combustion of an isolated droplet which is now well understood within the framework of quasi-steady, spherically symmetric diffusion theory. The classical theory of diffusion-controlled combustion of an isolated droplet in a quiescent oxidizing environment, yields the well-known d^2 -law. Under quasi-steady conditions, fuel vapor diffuses outward while oxidizer diffuses inward, and a spherical diffusion flame forms at a finite stand-off distance from the droplet surface. The burning rate is governed by the Spalding mass transfer number.

However, practical systems rarely involve droplets burning in isolation. Instead, droplets burn as clusters or dense clouds in which transport processes of individual droplets influence one another and the interactions between neighboring burning droplets cannot be ignored. When multiple droplets burn simultaneously, their vapor concentration and temperature fields overlap. The oxygen consumed by one droplet depletes the availability for the neighboring droplets. From the mathematical perspective of combustion theory, the presence of each droplet modifies the boundary conditions experienced by every other. This breaks down the spherical symmetry of an isolated droplet, resulting in the deformation of flame front of droplets, and eventual flame merging when droplets get sufficiently close together. As a result, the burn rate of a cluster of droplets can substantially differ from the sum of their individual burn rates. The group combustion of droplets therefore represents a central problem in multiphase reacting flows.

Several fundamental aspects of group burning are now clear. First, droplet interaction is funda-

mentally a diffusion-mediated phenomenon where the thermal and concentration perturbations of individual droplets overlap to modify gradients at droplet surfaces and thereby alter burn rates. Second, burning rate suppression increases with decreasing spacing and increasing number density, reflecting enhanced shielding of oxidizer transport. Third, geometric symmetry breaking leads to flame deformation and eventual merging, marking a transition from isolated to group combustion. Finally, in sufficiently dense systems, the relevant length scale governing combustion shifts from the droplet diameter to the overall cloud dimension.

One approach to improving the modeling framework is to develop a better closure for the burn rate of a droplet that takes into account the perturbing effect of all the neighbors. Such a multi-droplet modeling framework is quite complicated if we want to account for the influence of all the neighbors in a combined manner. A much simpler pairwise interaction approach has been advanced in the context of force modeling, where the perturbation effect of each neighbor is taken into account one at a time in a decoupled manner and then superposed to obtain the total influence of all the neighbors [Akiki et al., 2017, Balachandar et al., 2020, Siddani and Balachandar, 2023, Seyed-Ahmadi and Wachs, 2020, Siddani et al., 2024, Jbara et al., 2025]. The simplified approach thus requires the perturbing influence of only one neighbor captured as a binary model. Using order invariance approximation, the same binary model can be used for the nearest neighbor, second nearest neighbor, and so on [Moore and Balachandar, 2019].

One focus of this proposal is to develop such a neighbor-informed binary modeling framework for group combustion. With the pairwise interaction and order invariance approximations, we only need a binary model that accounts for the perturbation influence of one neighbor on another in a two droplet system. We closely follow the work of Li and Zhang [Li and Zhang, 2024] and use their bispherical numerical approach to solve the quasi-steady combination of two stationary droplets. Following [Li and Zhang, 2024] we make the assumptions of unit Lewis number, negligible cross flow, infinitely thin flame sheet, and potential flow.

The burn rate of a droplet in a two-droplet system depends on the droplet’s size and Spalding mass number, and neighbor’s location, size, and Spalding mass number. The first step we take is to substantially simplify the complete representation of this complex binary dependence into a few simple interaction functions. With these interaction functions we develop the ability to capture the perturbing influence of any neighbor in a two-droplet system. We then develop the pairwise superposition framework and use the complete representation of the binary interaction to study a large multi-droplet system. In addition to burn rate, we extend the pairwise superposition framework to modeling of convective heat transfer and force on a droplet due to the Stefan flow induced by the neighboring droplets.

For the two-interacting droplet problem, a simple unary model based on the explicit superposition of adding the solutions for two individual droplets will violate the boundary conditions; i.e., merely adding the two droplets and two surface spherical harmonic modes

$$\phi \approx \sum_{i=1}^2 \phi_i^{(I)}. \quad (2.34)$$

Therefore, a new unary implicit method was developed such that the superposed solution must satisfy the boundary conditions. One way to allow interaction between the droplets is to let

$$\phi^{(M+D)}(\mathbf{x}) \approx \alpha_1 G_1^{(0)}(\mathbf{x}) + \beta_1 G_1^{(1)}(\mathbf{x}) + \alpha_2 G_2^{(0)}(\mathbf{x}) + \beta_2 G_2^{(1)}(\mathbf{x}) \quad (2.35)$$

where $G_i^{(0)}$ is the general monopole solution and $G_i^{(1)}$ is the general dipole solution. Then satisfying the boundary conditions on each droplet surface results in solving the matrix equation

$$A_\phi^{(M+D)} \begin{bmatrix} \alpha_1 \\ \beta_1 \\ \alpha_2 \\ \beta_2 \end{bmatrix} = \begin{bmatrix} \phi_{s,1} \\ 0 \\ \phi_{s,2} \\ 0 \end{bmatrix}, \quad \text{where} \quad A_\phi^{(M+D)} = \begin{bmatrix} 1 & 0 & M_{12}^{00} & M_{12}^{01} \\ 0 & 1 & M_{12}^{10} & M_{12}^{11} \\ M_{21}^{00} & M_{21}^{01} & 1 & 0 \\ M_{21}^{10} & M_{21}^{11} & 0 & 1 \end{bmatrix} \quad (2.36)$$

The implicit monopole solution sets $G_i^{(1)} = 0$, while the implicit dipole solution sets $G_i^{(0)} = 0$. Figure 10 shows results for the quantities of interest: the total evaporation rate, the net axial force, and conductive heat transfer. Results indicate that the implicit monopole solutions and the implicit dipole solutions are reasonably good fits. Future work will consider the combined full expansion of monopole and dipole terms in equation (2.35). The goal of such modeling is to guide the ML modeling efforts in group combustion.

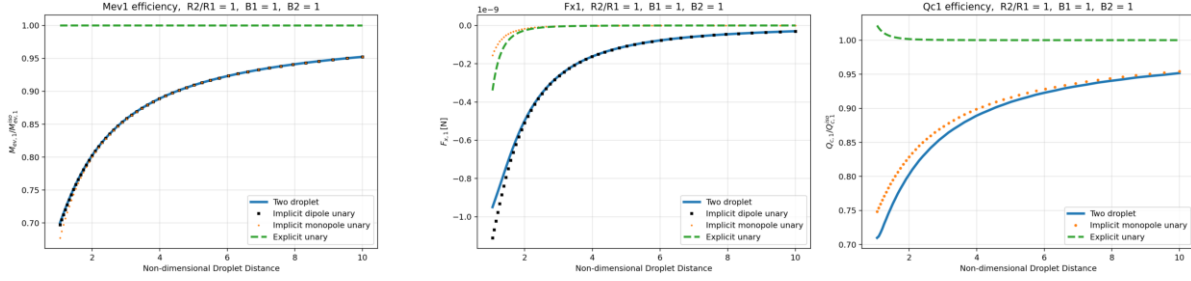


Figure 10: Comparison of models for the two-interacting droplet problem. Plots show evaporation rate (left), force (middle), and heat transfer (right). In all cases the implicit multi-pole superposition method works well.

2.3 What opportunities for training and professional development has the project provided?

The Center has provided the opportunity to train many graduate students, postdocs and undergraduate students in collaborative team-based research. In addition, the project has provided the Center with the opportunity to make a significant impact on the educational experience of leveraged PhD students at UF, Purdue, and GT. Leveraged students are those pursuing a PhD in a field so closely related to the topic of the Center that they contribute, and in turn learn, from the daily interactions, internal meetings, and research carried out by the Center. By combining the workforce of funded and leveraged students, significant progress has already been realized. The disciplines of gpu computing, fluid dynamics and multiphase flows, and machine learning, make it ripe for training the next-generation workforce and transfer to educational classroom.

2.4 How have the results been disseminated to communities of interest?

Results have been presented to the scientific community in terms of publication, conference papers, and talks. For a detailed list please see Sec. 3.

2.5 What do you plan to do during the next reporting period to accomplish the goals and objectives?

In order to meet the objectives of the proposed work, and to follow the simulation roadmap as well as the time table, the Center has been meeting weekly in cross-disciplinary teams. We consider these weekly meetings essential, and will be continuing them throughout the life time of the Center. During the next reporting period, the Center for Multiscale Modeling of Multiphase Combustion (CM³C) will focus on advancing all major components of the integrated multiscale framework, with emphasis on completing critical code capabilities, generating the first high-fidelity datasets, initiating machine-learning model training, and expanding validation experiments.

Recruitment and Team Development: We will continue recruiting outstanding graduate students and postdoctoral researchers, with particular emphasis on strengthening the machine learning, uncertainty quantification, and experimental teams. New personnel will be integrated into the existing cross-disciplinary team structure and will participate in weekly team meetings and Center-wide coordination meetings.

High-Fidelity Particle-Resolved Simulations: The microscale team will initiate implementation of reacting immersed boundary conditions for carbon particles, including heterogeneous surface chemistry and heat transfer. The new spatial filtering capability will be finalized and incorporated into MFC to compute and output all filtered quantities required for machine learning model development. Verification studies will be completed, followed by the first particle-resolved simulations of dynamic particles in high-speed flow to generate training data for closure model development.

Mesoscale and Macroscale Euler–Lagrange Simulations: The Euler–Lagrange team will complete implementation of two-way coupled force, heat transfer, and particle reaction models. Particle motion, collisions, and source-term projection will be further optimized and tested at large scale. We will carry out the first simulations of the hemispherical blast problem and of shock-driven particle curtains, providing baseline datasets for comparison with planned experiments.

MFC Development and Exascale Performance: Development of MFC will continue to ensure that all newly added multiphase and reactive capabilities maintain excellent GPU performance and scalability on leadership-class computing platforms. The code infrastructure for direct incorporation of machine-learning-based closure models will be developed. Additional testing and optimization will be performed on LLNL and DOE computing systems.

Machine Learning Model Development: The machine learning team will begin training generative neural network models using the first high-fidelity microscale datasets. In parallel, graph neural network and operator-learning architectures will be further developed to model subgrid stresses, interphase coupling terms, and heat-release closures. Multi-fidelity strategies that combine high-fidelity simulation data with lower-cost surrogate data will also be explored.

Group Combustion Modeling: The group combustion team will continue developing deterministic theories for neighbor-informed burning and identifying the key descriptors needed to characterize particle interactions. We will develop a fundamental theory for superposition of modular solutions. These theoretical advances will guide the design of closure models and help define physically meaningful features for machine learning.

Experimental Program: The experimental team will complete validation of the single carbon rod facility against published results and begin designing and constructing multi-rod configurations to study group combustion effects. Additional diagnostics and data-processing methods will

be developed to quantify reaction rates and uncertainty. Preparatory work will also continue on the hemispherical blast experiments, including design of instrumentation and blast-hardened diagnostics.

Uncertainty Quantification and Workflow Integration: The UQ and workflow team will finalize the uncertainty quantification for the microscale simulations and carbon rod experiments. Initial uncertainty propagation studies will be conducted to identify the dominant contributors to prediction uncertainty. Workflow tools for data transfer, storage, and post-processing across simulation, machine learning, and experimental teams will be refined.

Overall Objective for the Next Reporting Period: By the end of the next reporting period, we expect to have (i) progressed the core code capabilities required for reactive microscale and Euler–Lagrange simulations, (ii) generated the first high-fidelity particle-resolved datasets for compressible high-speed multiphase environment, (iii) initiated training of generative and operator-learning models with these relevant datasets, (iv) advanced towards exascale EL simulations of explosive dispersal of particles, and (v) produced the first validation-quality experimental datasets. These milestones will establish the first complete end-to-end demonstration of the Center’s multiscale, machine-learning-enabled predictive simulation framework.

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3 PRODUCTS

3.1 Publications, conference papers, and presentations

3.1.1 Publications

1. T. Daoud, T.L. Jackson, S. Balachandar. (2026) A Careful Examination of Closure Models in Euler-Lagrange Simulations of Compressible Multiphase Flow in a Planar Shock Particle Curtain Problem, *International Journal of Multiphase Flow*, **201**, 105740.
<https://doi.org/10.1016/j.ijmultiphaseflow.2026.105740>
2. Prosperetti, A., Lott, C. P., Balachandar, S. (2026) Force on a sphere in a general potential flow. *Journal of Fluid Mechanics*, **1034**, R3.
<https://doi.org/10.1017/jfm.2026.11504>
3. C.T.V. Nunes, C.D. Willhardt, J.L. Manin, E. Berrocal, D.R. Guildenbecher, Corrections for Accurate Measurements of Particle Extinction in Scattering Media using Collimated and Diffuse Illumination, *Optics Express*, **34**,R8, pp. 14068-14085
<https://doi.org/10.1364/OE.591768>

3.1.2 Conference Papers

1. C.T.V. Nunes, R.L. Travis, D.R. Guildenbecher (2026) Extinction Imaging of Particle-Laden Explosive Blast Waves, *Spring Technical Meeting of the Central States Section of the Combustion Institute*, West Lafayette, Indiana, May 17-19.
2. A.V. Millauer-Hsu, C.T.V. Nunes, D.R. Guildenbecher (2026) Measurements of Carbon Rod Combustion Rates, *Spring Technical Meeting of the Central States Section of the Combustion Institute*, West Lafayette, Indiana, May 17-19.
3. G. O'Brien, J.W. Posey, C. Spangler, R.W. Houim (2026) Modeling the Dispersal, Ignition, and Combustion of Magnesium Particles by an Explosive Charge, *Spring Technical Meeting of the Central States Section of the Combustion Institute*, West Lafayette, Indiana, May 17-19.

3.1.3 Presentations

- J. B. Harley, Physics to Filters: Understanding Nature Via Signal Processing, Invited Lecture, University of Florida IEEE Signal Processing Society Chapter, 11 March (2026).
- J. B. Harley, Artificial Intelligence with Physical Knowledge: Examples in Materials Science and Nondestructive Evaluation, University of Florida Fluids Seminar, 9 October (2025).
- S. Balachandar, Modeling and Machine Learning For Massive Simulations of Multiphase Flows, Invited Seminar, Mechanical and Aerospace Engineering Department, University of Houston, Houston, TX, 6 March (2026).
- S. Balachandar, Center for Multiscale Modeling of Multiphase Combustion (CM3C), Invited Lecture, DOE-NNSA PSAAP-4 Kickoff Meeting, 26 August (2025).
- S. Balachandar, Shock-Particle Interaction and Explosive Dispersal of Particles, Invited Seminar, Aerospace Engineering Department, Indian Institute of Technology Madras, Chennai, India, 21 January (2026).

- S. Balachandar, How to Harness the Powers of Machine Learning for Unprecedented Advances in Multiscale Modeling of Multiphase Flows, Invited Lecture, AI-Driven CFD Symposium, Gandhinagar, India, 13 January (2026).
- S. Balachandar, Investigation of the Structure and Dynamics of Particle-Laden Turbidity Currents, Invited Lecture, MADRAS-2026, Zanzibar, Tanzania, 7 January (2026).
- S. Balachandar, How to Harness the Powers of Machine Learning for Unprecedented Advances in Multiscale Modeling of Multiphase Flows, Invited Lecture, American-Japanese-European Two-Phase Flow Group Meeting, MIT, MA, 22 September (2025).

3.2 Website(s) or other Internet site(s)

The MFC and CM³C Slack channels are:

1. <https://mflowcode.slack.com>
2. <https://cm3c.slack.com>

The MFC website is: <https://mflowcode.github.io>, and the code is available in an MIT open source GitHub repository: <https://github.com/MFlowCode/MFC>.

3.3 Technologies or techniques

None to report.

3.4 Inventions, patent applications, and/or licenses

None to report.

3.5 Other products

None to report.

4 PARTICIPANTS & OTHER COLLABORATING ORGANIZATIONS

This section is optional. Since many of the items are discussed in other sections of the report, it is not duplicated here.

5 IMPACT

a. **What was the impact on the development of the principal discipline(s) of the project?**

The advancement of the science of group combustion is expected to be significant. Advancement will be in the forms of multi-rank gpu computing; machine learning, specifically operator and generative learning models that rely upon physics; and group combustion modeling, specifically understanding the close interaction of multiple reacting droplets in a compressible flow. Additionally, the models and algorithms developed as part of the Center will be publicly available through the MFC GitHub account.

b. **What was the impact on other disciplines?**

Although it is too early to gauge the impact of the Center’s research on other disciplines, it is expected that the novel and ground-breaking ideas merging machine learning and fluid mechanics/group combustion will translate into similar ideas in other physics or engineering areas that require machine learning. We expect advancements will be made in operator and generative machine learning, and these ideas will translate into class courses in computer-science related fields.

c. **What was the impact on the development of human resources?**

The center is training many graduate students, postdocs and undergraduate students in collaborative team-based research.

d. **What was the impact on teaching and educational experiences?**

The Center has made a significant impact on the educational experience of leveraged PhD students at UF, Purdue, and GT. Leveraged students are those pursuing a PhD in a field so closely related to the topic of the Center that they contribute, and in turn learn, from the daily interactions, internal meetings, and research carried out by the Center. By combining the workforce of funded and leveraged students, significant progress has already been realized. The disciplines of gpu computing, fluid dynamics and multiphase flows, and machine learning, make it ripe for training the next-generation workforce and transfer to educational classrooms.

e. **What was the impact on physical, institutional, and information resources that form infrastructure?**

Nothing to report.

f. **What was the impact on technology transfer?**

MFC is a publicly available code that can be downloaded from GitHub. As such, all models incorporated by the Center can be used, and improved upon, by other research groups in similar fields. Pull requests (PR) are done at periodic times, so that the MFC code is continually updated with the most recent advancements, and is then immediately available for download.

g. **What was the impact on society beyond science and technology?**

Although it is too early to gauge, many of the EL algorithms and models can be used in a wide range of disciplines, including medicines, volcano eruptions, aerospace, bio-engineering, etc.

h. **What percentage of the award’s budget was spent in foreign country(ies)?**

No funds were used in foreign countries.

6 CHANGES/PROBLEMS

a. **Changes in approach and reasons for change:**

To date, no changes have been made in the approach as originally outline in the proposal.

b. **Actual or anticipated problems or delays and actions or plans to resolve them:**

We do not anticipate problems or delays in the current or upcoming funded cycles.

c. **Changes that have a significant impact on expenditures:**

None to date.

d. **Significant changes in use or care of human subjects, vertebrate animals, biohazards, and/or select agents:**

The proposed work does not use, nor does the proposed work call for the use, of human subjects, vertebrate animals, biohazards, and/or select agents.

e. **Change of primary performance site location from that originally proposed:**

The location of the primary site location, the University of Florida, Gainesville, has not changed nor is it expected to change during the lifetime of the Center.

7 SPECIAL REPORTING REQUIREMENTS

a. Provide an update on recruitment plans, activities, and outcomes.

The Center was quickly staffed with truly outstanding PhD students and postdocs in all areas of the proposed work. These include students in fluid mechanics, combustion, machine learning, and computational science. These students are grouped into the following teams:

- (a) Microscale Team
- (b) Meso/Macroscale Team
- (c) MFC code development team
- (d) Experimental Team
- (e) Machine Learning Team
- (f) Group Combustion Team
- (g) UQ/Workflow Team

The postdocs oversee and participate with one or more teams, and are responsible for leading the weekly team meetings. The weekly team meetings are cross-populated with appropriate members from the other teams and faculty

b. List any students and postdocs supported by the project. Include name, position, and percentage of support from this award. For students, please also include their anticipated degree (including major and graduation date), advisor(s), and completed, planned, or anticipated Laboratory internship date.

Name	Position	Support	Degree/Date	Advisor	Internship
Joshua Gillis	PhD (UF)	100%	ME [2029]	Balachandar	Summer 2027
Isabella Rivera Sigmann	PhD (UF)	100%	ME [2029]	Balachandar	Summer 2027
BC Kim	Postdoc	100%			Summer 2027
Alex Millauer-Hsu	PhD (Purdue)	75%	ME [2030]	Guildenbecher	Summer 2028
Cohen Nunes	PhD (Purdue)	75%	ME [2027]	Guildenbecher	Summer 2027
Isaly Tappan	PhD (UF)	100%	ECE [2027]	Harley	Summer 2027
Erica Lindbeck	Postdoc	100%			Summer 2027
Juan Diaz	PhD (UF)	100%	ECE [2029]	Zare	Summer 2027
Jose Aguilar	Postdoc	100%	MAE [2026]	Balachandar	Summer 2027
Colton Willhardt	Postdoc	25%		Guildenbecher	
Gregory O'Brien	PhD (Purdue)	75%		Houim	
Jacob Posey	Postdoc	25%		Houim	
Benjamin Wilfong	PhD (GT)	25%	CSE/ME [2028]	Bryngelson	Summer 2026
Daniel J. Vickers	PhD (GT)	100%	CSE [2030]	Bryngelson	
Dimitrios Adam	PhD (GT)	50%	CSE/AE [2028]	Bryngelson	

c. Provide a brief description about each graduate student who has interned during the reporting period at one of the DOE/NNSA National Laboratories (at least 10 weeks) as required under the PSAAP IV Notice of Funding Opportunity. Please provide their name, the Laboratory name, Laboratory staff contact/mentor, length of time spent, and a brief explanation of the research or internship experience.

None have interned at the NNSA laboratories. However, we currently have the follow-

ing students that will be doing their internship in summer of 2026.

Name	Laboratory	Mentor	Length (wks)	Research
Thierry Daoud	Livermore	Dr. Georges Akiki	10	Pseudo-turbulence modeling connected with multiphase flows
Cohen Nunes	Sandia	Dr. Michael Kaneshige	10	Experimental Detonation Sciences

- d. **List any students that have graduated, and students, graduates, or postdocs that have been hired by the national labs or other government agencies and the name of the hiring institution.**

None to report.

8 BUDGETARY INFORMATION

Budgetary information is not required for semi-annual reports.

9 PROJECT OUTCOMES

The Center for Multiscale Modeling of Multiphase Combustion (CM³C) has established the scientific, computational, and organizational foundation needed to achieve its long-term goal of predictive simulation of phase change and group combustion in dense clouds of reactive particles subjected to high-speed and detonation environments. During the first six months of the project, the Center successfully integrated expertise in multiphase flow, combustion, machine learning, uncertainty quantification, and exascale computing across the University of Florida, Georgia Institute of Technology, and Purdue University, and recruited an outstanding team of graduate students and postdoctoral researchers organized into focused technical teams.

A major outcome of this reporting period is the rapid advancement of the Multicomponent Flow Code (MFC) into a unified exascale simulation platform capable of addressing the microscale, mesoscale, and macroscale problems central to the CM3C roadmap. New capabilities were implemented for particle-resolved simulations, including moving immersed boundaries, particle–particle collisions, periodic boundary conditions, and efficient output of filtered quantities required for machine learning. In parallel, substantial progress was made in developing Euler–Lagrange capabilities for tracking and coupling billions of reactive particles, including high-order interpolation, Gaussian projection methods, and state-of-the-art gas–particle force models. These developments position MFC to serve as the Center’s core computational engine for all planned simulations.

The project also established the data-generation and machine-learning workflow that will connect high-fidelity particle-resolved simulations to reduced-order closure models for large-scale simulations. The required filtered quantities and closure terms were identified, and the computational infrastructure was defined to generate training datasets for both generative neural networks and operator-learning models. This framework will enable the development of “best-possible” closure models that account for the influence of neighboring particles and provide a rigorous, physics-informed route for transferring microscale knowledge to mesoscale and macroscale simulations.

Complementary progress was made on the experimental program. Carbon rod experiments for validating heterogeneous reaction models were successfully established, and preliminary measurements of surface regression as a function of temperature were obtained. Planning and characterization also advanced for the hemispherical blast experiments that will provide validation data for mesoscale and macroscale simulations of explosively dispersed reactive particle clouds. These experiments, combined with the Center’s uncertainty quantification framework, will provide validation-quality data and quantitative confidence bounds for the predictive models developed by the Center.

Finally, the Center developed a rigorous theoretical and organizational framework for understanding and modeling group combustion. The team identified the key physical closure terms, established a coordinated uncertainty quantification and workflow strategy, and initiated theoretical efforts to deterministically account for neighbor-informed burning behavior. Collectively, these accomplishments provide a strong foundation for future years, when machine-learning-based closure models will be integrated into exascale simulations and validated against experiments to deliver predictive capability for high-speed multiphase combustion systems.